

Title page

Effects of perceptual salience and task load on executive vigilance: Independent and interacting effects without modulation of the vigilance decrement.

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Abstract page

Abstract

Objective: To examine the independent and interacting effects of perceptual salience and task load on executive vigilance performance, as well as whether these factors modulate the vigilance decrement over time.

Background: Hits in vigilance tasks typically decline with time-on-task. This decrement is explained by the resource depletion of attentional resources (overload model) or to disengagement caused by task-unrelated thoughts (underload model). Previous research manipulated salience or task load separately, yielding inconsistent findings. Moreover, few studies have examined both factors simultaneously while holding other task parameters constant.

Method: Two preregistered studies and a pooled data analysis were conducted using the ANTI-Vea task, which assesses attention and vigilance components. In Study 1 ($N = 84$), perceptual salience of the target stimulus was manipulated between participants. In Study 2 ($N = 45$), perceptual salience and task load were orthogonally manipulated within participants. Study 3 reanalyzed combined data ($N = 238$) from Studies 1 and 2 together with data from previous research.

Results: Executive vigilance showed significant effects of perceptual salience and task load, with larger overall accuracy observed for the most salient targets and under moderate task load (dual-task condition). In signal detection measures, A' was largest for the most salient targets, whereas moderate task load was associated with a less conservative response bias (B''). However, neither variable modulated the observed vigilance decrement over time.

Conclusion: Perceptual salience and task load affect overall executive vigilance but do not modulate its decrement over time-on-task.

Application: Findings may inform the design of tasks and systems that optimize vigilance by promoting an optimal balance of task demands, by considering perceptual salience and task load to minimize failures and reduce the risk of vigilance decrement-related accidents.

Keywords: Vigilance, Executive vigilance, Perceptual salience, Task load, Vigilance decrement

Précis: This study examined how perceptual salience and task load influence executive vigilance using the ANTI-Vea task. Across two preregistered studies and a pooled data analysis, overall performance varied across levels of task load and perceptual salience, but such manipulations did not modulate vigilance loss over time-on-task. Findings contribute to a fine-grained understanding of task-related factors that affect vigilance performance.

Introduction

The vigilance decrement is defined as a progressive loss in the ability to monitor the environment and detect rare but critical stimuli, as evidenced by an increase in erroneous and slower responses over time (Davies & Parasuraman, 1982). The vigilance decrement can lead to mild consequences, such as missing a reading, or severe ones, such as a traffic accident (Baysari et al., 2008; Edkins & Pollock, 1997). The study of vigilance has gained considerable interest in understanding the causes and mechanisms associated with the vigilance decrement and to optimize performance in prolonged tasks (Thomson et al., 2015). Several theoretical models have been proposed, which currently might be grouped into two relatively opposing frameworks: overload and underload theories (Thomson et al., 2015).

For decades, the most widely accepted explanation for the vigilance decrement has been the overload theory based on the resource depletion hypothesis (Warm et al., 2008). According to this view, attention relies on limited resources that are consumed during sustained activity and are not quickly replenished (Dillard et al., 2019; Grier et al., 2003). In general, the resource depletion hypothesis posits that vigilance tasks—because they require substantial mental effort—consume more resources and produce a greater vigilance decrement. However, recent research has suggested that these resources may be renewable (Helton & Wen, 2023).

In contrast, underload theories include proposals such as the mindlessness hypothesis (Robertson et al., 1997), the resource control hypothesis (Thomson et al., 2015), and the mind-wandering hypothesis (Smallwood & Schooler, 2006). In particular, the mind-wandering hypothesis, unlike the resource depletion account, argues that the vigilance decrement does not result from a depletion of resources, but rather from the redistribution of

98 attentional resources toward irrelevant internal thoughts (i.e., mind-wandering). From mind-
99 wandering hypothesis, vigilance tasks are considered undemanding, monotonous, and
100 cognitively unstimulating, which facilitates attentional disengagement (Neigel et al., 2020).
101 Consequently, the mind-wandering hypothesis predicts that more demanding or more
102 engaging tasks should reduce the likelihood of mind-wandering and, in turn, mitigate the
103 vigilance decrement (Smallwood, 2010), whereas boring or low-demand tasks should
104 promote a faster withdrawal of attention and a more pronounced decline in performance over
105 time (Thomson et al., 2015).

106 Numerous empirical studies have tested the assumptions of the resource depletion and
107 mind-wandering hypotheses by manipulating task characteristics, such as load and/or
108 physical features of stimuli, with the latter often operationalized through perceptual salience
109 (Esterman & Rothlein, 2019; Gonçalves et al., 2017; Helton & Warm, 2008; Seli et al., 2018).
110 However, it appears that both theories are currently unable to explain the reported findings
111 (Esterman & Rothlein, 2019; Neigel et al., 2020; Thomson et al., 2015).

112 In particular, task load is manipulated by varying the number of tasks performed
113 simultaneously and the number of instructions that need to be maintained. Several studies
114 show that performing a dual task affects vigilance more than performing a single task. For
115 example, Head and Helton (2014) found that hits in the Sustained Attention to Response Task
116 (SART) were worse when combined with a verbal memory task, compared to performing the
117 SART alone. Epling et al. (2016) also observed a greater vigilance decrement when a
118 semantic vigilance task was combined with a word recall task. Other studies have
119 demonstrated that working memory load negatively affects vigilance (Helton and Russell,
120 2011). Smit et al. (2004) found that performance in a vigilance task was worse when

participants had to detect two stimuli instead of one. Overall, these results support the resource depletion theory, which proposes that higher cognitive demands deplete limited resources more quickly.

However, other studies indicate that increasing task load do not necessarily impair target's detection in the vigilance task and, in some cases, may mitigate the vigilance decrement. In this regard, for example, a correlational pattern was observed by Seli et al. (2016), who found that unintentional mind-wandering was more frequent during more demanding tasks, whereas intentional mind-wandering was more common during easier ones. Building on this, Seli et al. (2018) tested the mind-wandering hypothesis by experimentally manipulating task difficulty: including a working memory component in the difficult condition and using a simple vigilance task in the easy one. Participants' responses were more accurate and faster in the easy condition, confirming the effectiveness of the manipulation. These results suggest that tasks with lower cognitive demands support more stable vigilance performance. Although the vigilance decrement was not directly measured, the authors examined in intentional and unintentional mind-wandering as a function of task difficulty. Overall, mind-wandering rates were lower in the easy task, though more often intentional—likely because the task required less cognitive control. In contrast, unintentional mind-wandering was associated with reduced performance in the difficult task. Also, Epling et al. (2019) found that vigilance remained stable when a semantic vigilance task was combined with a narrative memory task. Similarly, in air traffic control tasks, performing two tasks simultaneously did not reduce vigilance compared to performing only one (Pop et al., 2012; Stearman and Durso, 2016; see also Luna, Barttfeld, et al., 2022). These results seem to

143 contradict the resource depletion hypothesis and could be better explained by the mind-
144 wandering hypothesis.

145 The lack of consensus in the results on the effects of task load on vigilance
146 complicates agreement on theoretical explanations for vigilance decrement. While some
147 findings align with the resource depletion hypothesis, others are better accounted for by the
148 mind-wandering hypothesis.

149 Alternatively, task demands may be modulated by perceptual salience of the target
150 stimuli. Evidence shows that more difficult tasks tend to yield worse overall performance and
151 a greater vigilance decrement. Gómez Iníguez et al. (2000) observed that in a difficult task
152 (with more distractors), there were fewer hits and more false alarms. Similarly, Helton and
153 Warm (2008) found that tasks with higher perceptual salience, i.e., easier tasks, led to reduced
154 vigilance decrement. These findings support the resource depletion theory, which suggests
155 that perceptually more demanding tasks require more resources, depleting them more quickly.
156 In general, these show that lower perceptual salience increases processing demands and tends
157 to exacerbate the vigilance decrement.

158 However, it has also been found that tasks with more salient targets can generate a
159 greater vigilance decrement. Ralph et al. (2017) compared two vigilance tasks with different
160 levels of demands: an easy task, the Mackworth Clock Test (MCT, Mackworth, 1948), where
161 participants had to detect repetitive stimuli in a monotonous sequence, and a difficult task
162 (Car Task), which simulated driving a car, requiring participants to stay in lane while
163 avoiding obstacles and compensating for wind gusts. Although the Car Task was perceived as
164 more demanding, a vigilance decrement only occurred in the easy task. This supports the

mind-wandering theory, as more engaging and demanding tasks seem to reduce task-unrelated thoughts and help maintain performance.

It is also important to note that vigilance theories have been developed using simple experimental tasks, such as the Sustained Attention to Response Task (SART) or the MCT. Although these paradigms allow for precise experimental control, their extremely simple and monotonous nature limits their ability to capture how vigilance unfolds in more dynamic or realistic contexts (Thomson et al., 2016). In an attempt to overcome these limitations, Szalma et al. (2014) developed a video game–based vigilance task that incorporated continuous movement through a complex and perceptually rich virtual environment. This approach represented an improvement over tasks such as the SART and the MCT by offering greater ecological validity and a less artificial context. However, this same characteristic also introduced an important limitation: although the task reproduced subjective indicators typically associated with vigilance (e.g., increased distress and reduced engagement), the vigilance decrement was not consistently observed, likely because the richer and more engaging environment reduced the monotony that usually triggers such deterioration. Moreover, simple vigilance tasks do not include additional measures of other attentional or vigilance components at the same time (Tkachenko & Dinges, 2018). Furthermore, when task load is manipulated, researchers often rely on different tasks (Epling et al., 2019; Head & Helton, 2014; Ralph et al., 2017) that vary in the variables assessed, the duration of the trials, and the characteristics of the stimuli. Consequently, the observed variations are difficult to compare among them, as task load is manipulated via different task parameters across studies.

A new tool for measuring multiple attentional and vigilance components in a single session has been developed by our lab: the ANTI-Vea task (Attentional Networks Test for Interactions and Vigilance – executive and arousal components; Luna et al., 2018; <https://anti-vea.ugr.es/>). The task captures the decrement on two vigilance components: (a) executive vigilance (EV), as the ability to monitor the environment to detect infrequent but critical signals, requiring higher-order cognitive processing such as updating working memory, making decisions, executing accurate responses to targets, and inhibiting responses to non-targets according to task goals (Hemmerich et al., 2025); and (b) arousal vigilance (AV), as the ability to maintain a fast response to an intermittent stimulus that always requires a response, which requires less cognitive control and facilitates a more automatic performance.

Importantly for our goals in the current study, the ANTI-Vea can be understood as a triple task that evaluates vigilance decrement under certain task load. In particular, this task integrates three type of trials, requiring different response sets: (a) the ANTI task, in which participants need to respond to the direction of a central arrow, surrounded by two more arrows on each side, and allows measuring the main effects and interactions between phasic alertness, orienting, and cognitive control (Petersen & Posner, 2012); (b) a signal detection task, similar to the SART, to assess the EV – by detecting an infrequent vertical shift of the central target arrow (present in 20% of the trials); and (c) a simple reaction time task, similar to the PVT, to measure AV in which a millisecond counter appears on the screen and participants must stop it as quickly as possible by pressing any key.

Several studies have replicated the validity and reliability of the ANTI-Vea across different experimental contexts (e.g., Coll-Martín et al., 2021; Luna, Roca et al., 2021),

including online versions of the task (<https://anti-vea.ugr.es/>). Critically, these investigations have shown that measures obtained in online sessions outside the laboratory are comparable to those obtained under standard laboratory conditions. In both formats, the EV decrement is reflected by a reduction in hits over time.

Considering the advantages of the ANTI-Vea task, Luna, Barttfeld et al. (2022) manipulated task load in the ANTI-Vea task by varying the number of tasks to be performed simultaneously. They found that the dual-task condition, in which participants only had to respond to the EV and AV trials, mitigated the EV decrement, maintaining hit rates stable over time, compared to the single-task (i.e., when they only had to respond to EV trials) and triple-task conditions, where the typical EV decrement was observed. However, task load did not affect AV.

The present study aimed to examine the effects of perceptual salience and task load on the vigilance decrement by using a task paradigm that enables both factors to be manipulated within a single, integrated task. By employing the ANTI-Vea, which comprises several subtasks performed under identical task parameters, perceptual salience and task load were varied by manipulating only a single task parameter for each factor.

To this end, we conducted two studies and a reanalysis of a pooled dataset. Studies 1 and 2 were pre-registered on OSF, including objectives, hypotheses, procedure and design, sample size estimation, and data analysis plan. Note that, however, these pre-registrations included hypotheses and analyses regarding additional dependent variables collected through the ANTI-Vea task that are not part of the main goal of the present study. Here, we will only report measures collected through the EV sub-task of the ANTI-Vea, as it is the only sub-task in which perceptual salience and task load were manipulated across studies.

In Study 1, the perceptual salience of the target (i.e., magnitude of the vertical displacement to be detected) was manipulated between participants at three levels, to examine whether performance (hits, FA, sensitivity and response bias) over time benefits or rather is hindered by different levels of target salience (Gómez Iníguez et al., 2000; Helton & Warm, 2008; Seli et al., 2018). In Study 2, participants completed nine sessions of the online ANTI-Vea (Luna, Roca et al., 2021) in which task load and target salience were manipulated orthogonally. In each session, task load was manipulated through instructions to complete one, two, or three of the ANTI-Vea subtasks, as in Luna, Barttfeld et al. (2022), and target salience was also varied across sessions, at three levels as in Study 1.

Study 3 reports a re-analysis of pooled data collected in Studies 1 and 2 of the present work together with data from a previous study on task load effects on vigilance (Luna, Barttfeld, et al., 2022). This re-analysis will enable us to further examine the effects of perceptual salience and task load on vigilance through data from a large sample size ($N = 238$).

Study 1

The present study aimed to examine the EV decrement via the online ANTI-Vea task as a function of different levels of perceptual salience of the target, to test some of the predictions of vigilance decrement models. Participants were assigned to one of three groups according to the perceptual salience of the target (i.e., low, standard, or high). Predictions were opposite for each theoretical account. According to the resource depletion hypothesis, it was expected that the lowest level of perceptual salience (i.e., highest difficulty to detect the target) would increase task demands and therefore hinder overall and time-on-task EV performance (hits, FA, sensitivity and response bias). In contrast, following mind-wandering

hypothesis it might be expected that a high level of perceptual salience would reduce task demands, therefore facilitating task disengagement. Consequently, a perceptually less demanding task would lead to a larger EV decrement over time.

This study was pre-registered on OSF (<https://osf.io/v3j5s/wiki/home/>). Although preregistered, data of cognitive control and AV measures exceed the aim of this study and therefore are presented in the Supplementary Material.

Method

Participants

In a previous study conducted with the ANTI-Vea (Luna, Barttfeld, et al., 2022), a summary analysis between experiments showed a between-within interaction with an effect size of $\eta_p^2 = .06$ for the change in hits (dependent variable) between task load (between-participant factor with three levels: single/dual/triple) and block (within-participant factor with six levels). Using G*Power 3.1.9.7 (Faul et al., 2007), the minimum sample size required to observe a similar effect size to the one reported in Luna et al. (2022) was a-priori estimated for the present study, in which a similar design was used (between-participants factor: perceptual salience with three levels, low/standard/high; within-participant factor: block with six levels). Considering a power of $1 - \beta = .95$ and an α error = .05, the minimum sample size required to observe an interaction between perceptual salience and block with an effect size of $\eta_p^2 = .06$ was $N = 84$. Anticipating the potential need to exclude some participants from data analysis (following data filtering for standard analyses conducted with the ANTI-Vea, see Luna et al., 2021), we decided to gather data from 90 participants ($n = 30$ by group).

Ninety healthy young adults (61 women; age: $M = 27.74$, $SD = 5.16$), who were undergraduate students from the City of Córdoba, Argentina, volunteered to participate in the present study. All of them had normal or corrected-to-normal vision. Participants completed a written informed consent on the LimeSurvey platform before data collection. The study was conducted according to the ethical standards of the 1964 Declaration of Helsinki (last update: Brazil, 2013) and was part of a larger research project approved by a local Ethical Committee (CEIIPsi; project code: PE 41 v.2). Participants were randomly assigned to one of three experimental groups according to perceptual salience (low/standard/high) of the EV sub-task in the ANTI-Vea. Due to technical issues, data from 6 participants could not be included in the data analysis. Thus, the final sample was of 84 participants (i.e., as a-priori estimated), who were part of the experimental groups as follows: $n = 28$ in the low, $n = 30$ in the standard, and $n = 26$ in the high perceptual salience task group.

Procedure and Design

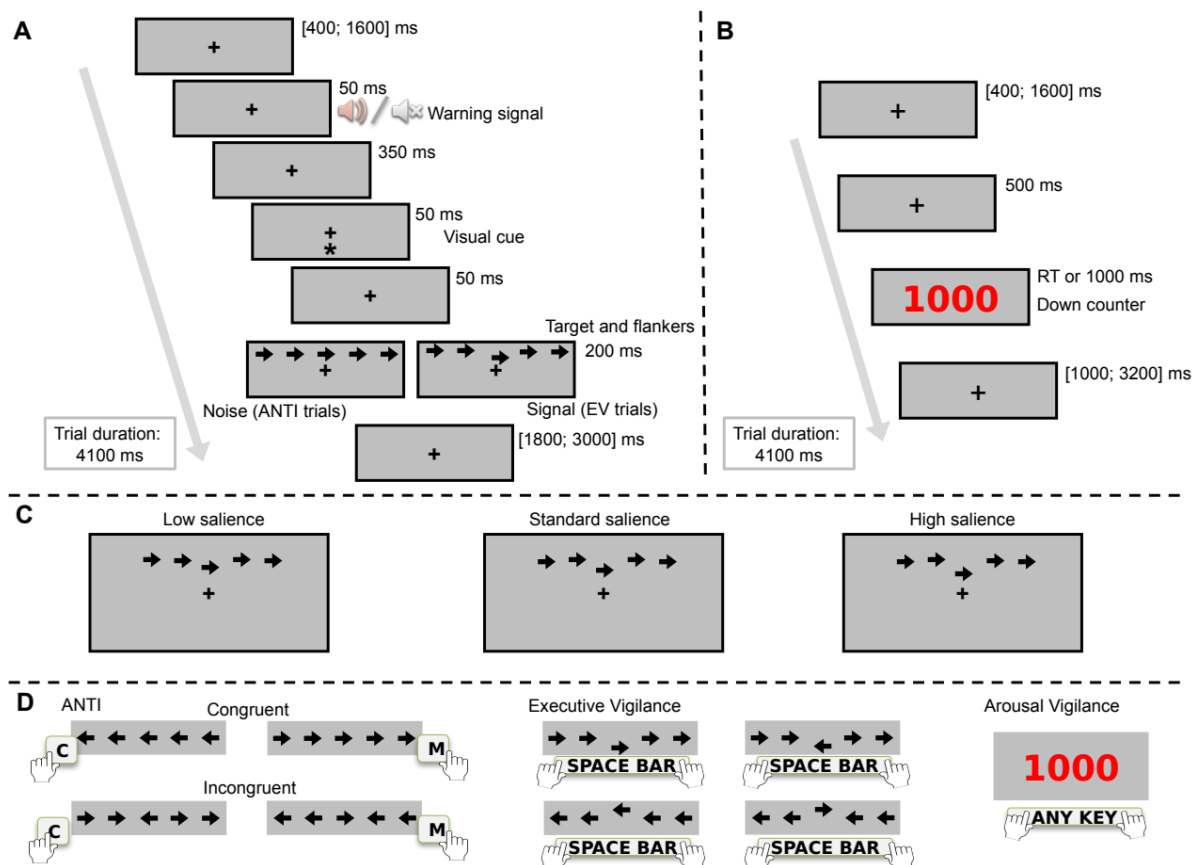
Participants received three web links via email to complete the procedure of the study in the following order: (a) the informed consent and a series of self-reported questionnaires in LimeSurvey (i.e., the Spanish version of the Mind Wandering-Deliberated/Mind Wandering-Spontaneous scales, the Discrete Emotional Scale [DES], and the Self-Assessment Manikin [SAM]); (b) the online ANTI-Vea task (<https://anti-vea.ugr.es/>), with instructions on how to select the perceptual salience of the EV sub-task according to their experimental group; and (c) another series of self-reported questionnaires on LimeSurvey to be completed post-task (i.e., NASA-TLX, thought identification question, DES, and SAM). With exception of the NASA-TLX, the inclusion of the remaining aforementioned self-reported questionnaires is, however, part of larger research project, and will be reported elsewhere.

301 ***Behavioral Task: ANTI-Vea***

302 The ANTI-Vea comprises three embedded sub-tasks, which are randomly completed
303 across three type of trials: (a) ANTI, a flanker task with an auditory warning signal and an
304 exogenous visual cue, suitable to assess phasic alertness, orienting, and executive control as
305 in the ANTI task of Callejas et al. (2004); (b) EV, a signal-detection task like the Mackworth
306 Clock Test (Mackworth, 1948) suitable to assess the EV decrement; and (c) AV, a RT task that
307 mimics the PVT (Lim & Dinges, 2008), suitable to measure the AV decrement. The
308 procedure for each type of trial is depicted in Figure 1 and explained in further detail in
309 previous studies and the online page of the task (<https://anti-vea.ugr.es/>, Luna et al., 2018;
310 Luna, Roca, et al., 2021).

311 **Figure 1**

312 *Procedure of the ANTI-Vea trials.*



313 *Note.* Procedure and timing for (A) ANTI and executive vigilance trials, and (B) arousal

314 vigilance trials. Note that, in all trials, responses were allowed until 2000 ms from target

315 onset. (C) A representative illustration of the perceptual salience between-participant

316 manipulation in the EV trials. (D) Correct responses for each type of trial.

317 In short, participants are instructed to fix their gaze on the fixation point at all times.

318 In the ANTI trials (60%, see Figure 1A), participants must select the direction pointed by the

319 target (i.e., the central arrow), while ignoring the direction pointed by the surrounding arrows

320 (see Figure 1D). Executive control is measured in the ANTI trials as the interference of

321 distractors in target selection between congruent (50% of ANTI trials) and incongruent (50%

of ANTI trials) see Figure 1D). Importantly, the ANTI trials also serve as the noise trials of the embedded signal-detection sub-task to assess EV. That is, if participants press the space bar key (i.e., the correct response to detect the infrequent signal in EV trials, see Figure 1D) as a response in an ANTI trial, the response is then categorized as a false alarm (FA). The EV trials (20%) follow the same procedure as the ANTI ones, except that the target is vertically displaced from its central position (see Figure 1A). Participants are instructed to detect the infrequent displacement by pressing the space bar, in these cases, ignoring the direction indicated by the target (see Figure 1D). Lastly, in AV trials (20%), no warning signal nor visual cue is presented (see Figure 1B). Participants are instructed to stop the millisecond counter as fast as possible by pressing any key from the keyboard (see Figure 1D).

The perceptual salience of the EV sub-task was manipulated between-participants (see Figure 1C). In the standard salience task, the vertical displacement of the target from its central position in the EV trials was fixed at 8 pixels (px), as in previous studies with the ANTI-Vea (Luna et al., 2018; Luna, Roca, et al., 2021). Instead, in the low salience task, the vertical displacement was fixed at 5 px, while in the high salience task was fixed at 14 px. Importantly, to increase the difficulty of detecting the target displacement in EV trials, small random displacements (± 2 px, applied to both the horizontal and vertical axes) were added to the position of stimuli. These displacements affected both the target and distractors in ANTI trials, but only the distractors in EV trials, wherein the target was always vertically displaced at its predefined level.

The experimental procedure comprised six consecutive blocks without breaks of 80 randomly presented trials (48 ANTI, 16 EV, and 16 AV), thus measuring vigilance in a period of 32 min 48 sec. Before experimental blocks, participants completed several practice blocks

345 with and without visual feedback, as in previous studies with the ANTI-Vea (Luna et al.,
 346 2018; Luna, Roca, et al., 2021).

347 ***Mental Workload: NASA-TLX***

348 After performing the ANTI-Vea, participants completed the Spanish translation
 349 (Arquer & Nogareda, 1999) of the short version of the NASA-TLX scale (Nygren, 1991).
 350 The short NASA-TLX assesses the mental workload perceived about the task performed
 351 across six items (i.e., mental, physical, and temporal demands, effort required, perceived
 352 performance, and level of frustration) that are rated on a scale between 0 and 20. A summary
 353 score of mental workload is obtained by summing the ratings of the six items. The NASA-
 354 TLX has shown acceptable reliability scores (split-half and Cronbach's α coefficients $> .80$;
 355 Xiao et al., 2005), and has been extensively used to assess mental workload (Grier, 2015;
 356 Hart, 2006), and specifically also widely used to assess the perceived mental workload
 357 associated with the vigilance decrement (Grier et al., 2003; Helton & Warm, 2008; Warm et
 358 al., 2008).

359 **Data Analysis**

360 Analyses were conducted using R 4.2.0 (R Core Team, 2024) in RStudio 2022.02.3
 361 (Posit team, 2024). Data was pre-processed with the reshape (Wickham, 2007) and plyr
 362 (Wickham, 2011) packages. Analysis of Variance (ANOVA) were conducted with the afex
 363 (Singmann et al., 2023) and emmeans (Lenth, 2023) packages. Effect sizes were computed
 364 with the effectsize package (Ben-Shachar et al., 2020) and CI following the Cousineau-
 365 Morey method (Morey, 2008) were calculated with the hausekeep (Lin, 2019) package.
 366 Figures were created using Matplotlib (Hunter, 2007).

The EV decrement was analyzed via standard analyses usually conducted with the ANTI-Vea (Luna, Barttfeld, et al., 2021). In all analyses, data were averaged by block and the perceptual salience of the task was treated as a between-participants factor with three levels (low/standard/high). Hits were computed as the correct responses in EV trials and FA as the space bar responses in ANTI (i.e., noise) trials, as in Luna, Barttfeld, et al. (2021). Signal detection theory metrics were computed as A' for sensitivity and B'' for response bias (Grier, 1971; Stanislaw & Todorov, 1999; Thomson et al., 2016). Polynomial contrasts of blocks were conducted to analyze trends of changes across time-on-task.

To identify potential outliers, we checked whether any participant exhibited a false alarm rate exceeding 50%; none met this criterion.

The mental workload score, obtained via the NASA-TLX scale, was analyzed through a one-way ANOVA, with perceptual salience of the task as between-participant factor. One participant was excluded from this analysis due to technical issues resulting in incomplete data.

Results

Table 1 shows average scores for EV measures as a function of the perceptual salience.

Table 1

Overall scores for EV as a function of the target's perceptual salience.

Dependent variable	Low salience		Standard salience		High salience	
	M	95% CI	M	95% CI	M	95% CI
Hit (%)	72.32	[67.13,	77.70	[72.70,	91.99	[86.60,

		77.51]	82.72]		97.37]	
False alarms	14.29	[11.50,	6.80	[4.10, 9.49]	1.59	[0.00, 4.48]
(%)		17.07]				
Sensitivity (A')	.87	[.86, .89]	.92	[.91, .93]	.98	[.96, .99]
Response bias	.29	[.11, .46]	.40	[.24, .57]	.20	[.02, .37]
(B'')						

Note. M = mean, 95% CI = 95% CI of the mean were computed following the Cousineau-Morey method (Morey, 2008).

Hits

The main effect of perceptual salience was significant (see Table 1 and Table 2). The largest percentage of hits was observed in the high salience task and was significantly different to that in the standard salience, $t(81) = 3.86$, $p < .001$, $d = 0.86$, 95% CI [0.40, 1.31], and in the low salience task, $t(81) = 5.23$, $p < .001$, $d = 1.16$ [0.69, 1.63], which did not differ from each other, $t(81) = 1.49$, $p = .303$, $d = 0.33$ [-0.11, 0.77].

Table 2

Repeated-measures ANOVAs for executive vigilance measures.

DV	Main effect or interaction	F	df	p	η^2_p [95% CI]
Hits	Perceptual salience	14.53	2, 81	< .001	.26 [.11, .41]
	Block	7.68	4.45, 360.10	< .001	.09 [.03, .13]
	Perceptual salience *	2.09	8.89, 360.10	.03	.05 [.00, .07]
	Block				
False	Perceptual salience	20.12	2, 81	< .001	.33 [.17, .47]

alarms	Block	6.6	4.58, 370.99	< .001	.08 [.03, .12]
	Perceptual salience *	1.92	9.16, 370.99	.047	.05 [.00, .07]
	Block				
Sensiti-	Perceptual salience	47.33	2, 81	< .001	.54 [.39, .65]
vity (A')	Block	0.3	4.61, 373.75	.903	< .01 [.00, .01]
	Perceptual salience *	0.26	9.23, 373.75	.986	< .01 [.00, <
	Block				.01]
Response	Perceptual salience	1.47	2, 81	.236	.04 [.00, .13]
bias (B'')	Block	4.98	4.39, 355.21	< .001	.06 [.01, .10]
	Perceptual salience *	1.94	8.77, 355.21	.047	.05 [.00, .07]
	Block				

Note. Significant outcomes are highlighted in bold. Abbreviations: df, degrees of freedom, DV, Dependent Variable.

As usually observed in the ANTI-Vea (Luna et al., 2018; Luna, Roca, et al., 2021), the decrement in hits was observed as a significant change across blocks (see Table 2), with a significant and negative linear component, $t(81) = -5.09, p < .001, \eta_p^2 = .24 [.10, .39]$ (see Figure 2A). Importantly, the change in hits across blocks was significantly modulated by the perceptual salience of the task (see Figure 2A). The linear component was significant in the low, $t(81) = -4.29, p < .001, \eta_p^2 = .19 [.06, .33]$, and in the standard, $t(81) = -3.63, p < .001, \eta_p^2 = .14 [.03, .28]$ salience task, but not in the high salience one, $t(81) = -1.01, p = .316, \eta_p^2 = .01 [.00, .10]$. Planned comparisons show that the linear decrement was significantly different between the low and high salience tasks, $t(81) = -2.25, p = .027, \eta_p^2 = .06 [.00, .18]$, but not

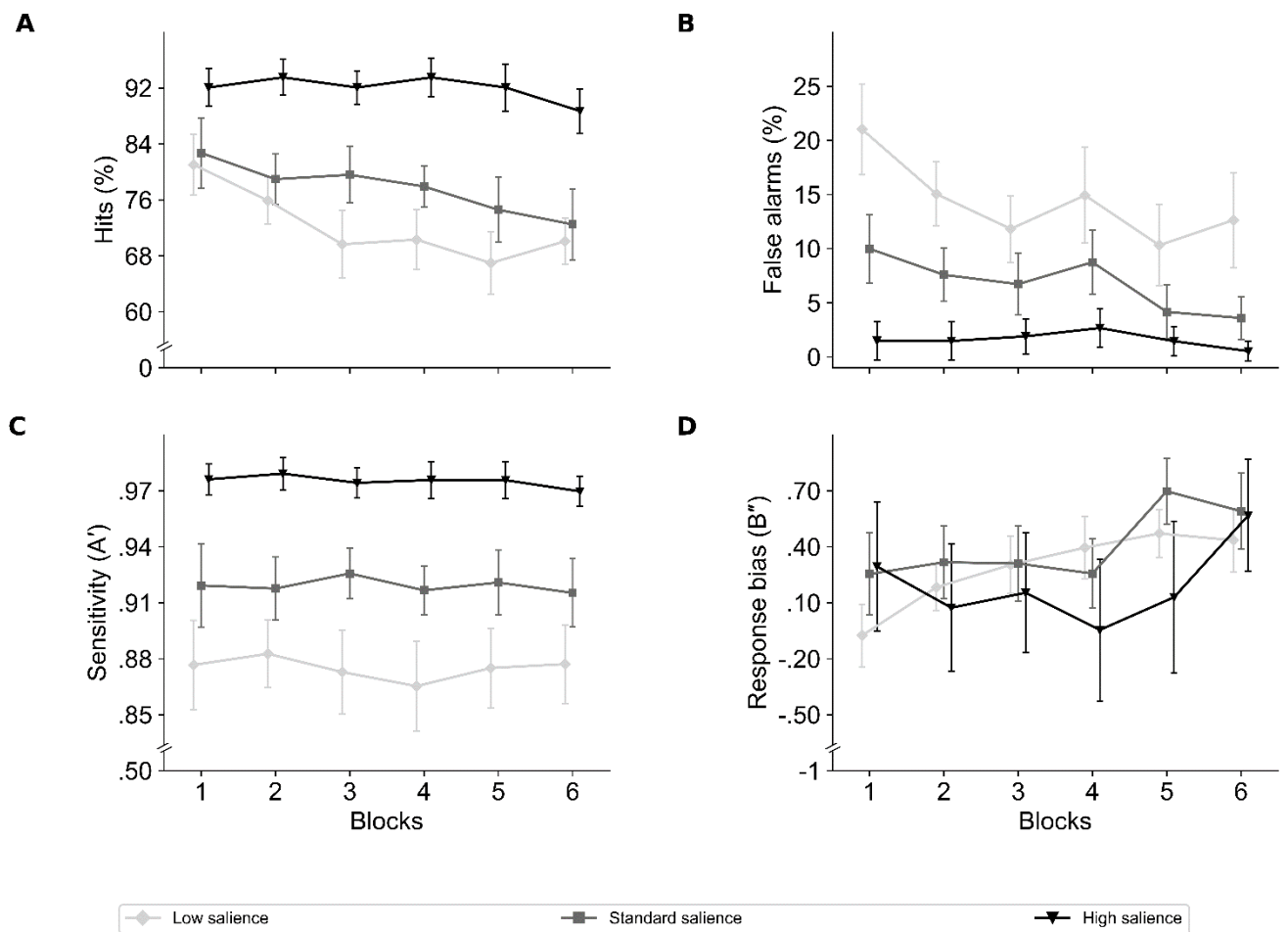
407 between the low and standard salience tasks, $t(81) = -0.57, p = .569, \eta_p^2 < .01 [.00, .07]$, and

408 between the standard and high salience tasks, $t(81) = -1.73, p = .088, \eta_p^2 = .04 [.00, .14]$.

409 **Figure 2**

410 *Executive vigilance performance across time-on-task as a function of the perceptual salience*

411 *of the task.*



412

413

414 *Note.* Error bars represent 95% CI of the mean and were computed following the Cousineau-

415 Morey method (Morey, 2008).

416 **False Alarms**

The main effect of perceptual salience on FA was significant (see Table 1 and Table 2). The percentage of FA was significantly larger in the low than in the standard salience task, $t(81) = 3.85, p < .001, d = 0.85 [0.40, 1.31]$; in the low than in the high salience task, $t(81) = 6.29, p < .001, d = 1.40 [0.91, 1.88]$; and in the standard than in the high salience task, $t(81) = 2.62, p = .028, d = 0.58 [0.14, 1.03]$.

As usually observed in the ANTI-Vea (Luna et al., 2018; Luna, Roca, et al., 2021), the decrement in FA was observed as a significant change across blocks (see Table 2) with a significant and negative linear component, $t(81) = -4.13, p < .001, \eta_p^2 = .17 [.05, .32]$ (see Figure 2B). Critically, as depicted in Figure 2B, the change in FA across blocks was significantly modulated by the perceptual salience of the task (see Table 2). Similarly to the pattern observed in hits, the linear component was significant in the low, $t(81) = -3.90, p < .001, \eta_p^2 = .16 [.04, .30]$, and in the standard, $t(81) = -3.07, p = .003, \eta_p^2 = .10 [.01, .24]$, salience task, but not in the high one, $t(81) = -0.29, p = .771, \eta_p^2 < .01 [.00, .05]$. Planned comparisons showed that the linear decrement in FA was significantly different between the low and high salience tasks, $t(81) = -2.47, p < .001, \eta_p^2 = .10 [.01, .23]$, but not between the low and standard salience tasks, $t(81) = -0.68, p = .502, \eta_p^2 < .01 [.00, .08]$, and between the standard and high salience tasks, $t(81) = -1.86, p = .064, \eta_p^2 = .04 [.00, .15]$.

Sensitivity

The main effect of perceptual salience on A' was significant (see Table 1 and Table 2). Overall A' was significantly larger in the high than in the standard salience task, $t(81) = 5.51, p < .001, d = 1.23 [0.75, 1.70]$; in the high than in the low salience task, $t(81) = 9.72, p < .001, d = 2.16 [1.61, 2.70]$; and in the standard than in the low salience task, $t(81) = 4.45, p$

439 $< .001$, $d = 0.99$ [0.53, 1.45]. Importantly, as depicted in Figure 2C, the change in A' across
 440 blocks was not significant (see Table 2), nor was it significantly modulated by the perceptual
 441 salience of the task (see Table 2).

442 **Response Bias**

443 The main effect of perceptual salience on B'' was not observed as significant (see
 444 Table 1 and Table 2). As usually observed in the ANTI-Vea (Luna, Barttfeld, et al., 2021;
 445 Luna et al., 2018; Luna, Roca, et al., 2021), a significant change in B'' across blocks towards
 446 a more conservative criterion was observed (see Table 2), with a significant and positive
 447 linear component, $t(81) = 4.21$, $p < .001$, $\eta_p^2 = .18$ [.05, .33] (see Figure 2D). The interaction
 448 between perceptual salience and blocks was observed as significant (see Table 2). The linear
 449 component was significant in the low, $t(81) = 3.36$, $p = .001$, $\eta_p^2 = .12$ [.02, .26], and in the
 450 standard, $t(81) = 2.75$, $p = .007$, $\eta_p^2 = .09$ [.01, .22], salience task, but not in the high-salience
 451 one, $t(81) = 1.24$, $p = .219$, $\eta_p^2 = .02$ [.00, .11], which instead showed a significant quadratic
 452 change across blocks, $t(81) = 3.51$, $p < .001$, $\eta_p^2 = .13$ [.03, .27]. Pairwise comparisons did not
 453 show significant differences in the linear component between tasks (all p 's $> .110$). A
 454 significant difference in the quadratic component was found between the low and high
 455 salience conditions, $t(81) = -3.66$, $p < .001$, $\eta_p^2 = .14$ [.03, .29], but not between the standard
 456 and high conditions, $t(81) = -1.54$, $p = .127$, $\eta_p^2 = .03$ [.00, .13].

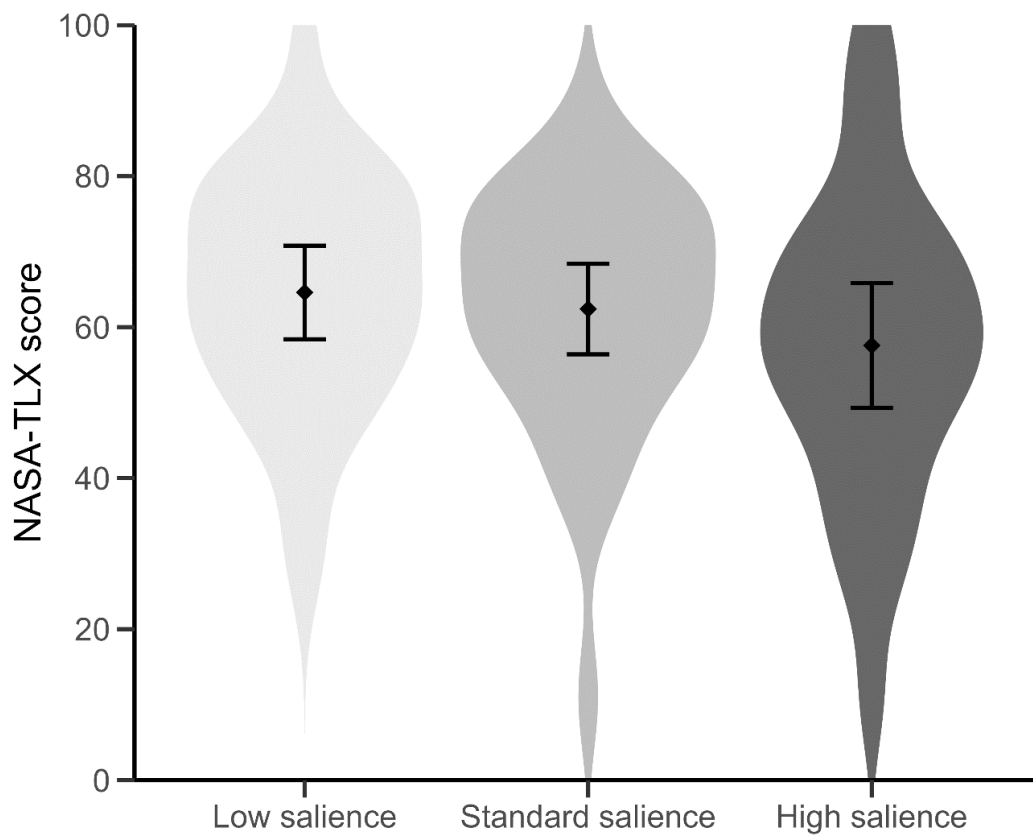
457 **Mental Workload: NASA TLX**

458 The main effect of signal salience was not significant for the global score of the
 459 NASA-TLX, $F(2, 80) = 1.12$, $p = .331$, $\eta_p^2 = .03$ [.00, .12]. As depicted in Figure 3, contrary

to the predictions by the resource depletion theory, participants reported a similar mental workload after completing the same vigilance task but with different signal salience.

Figure 3

Mental workload (mean NASA-TLX score) as a function of the signal salience of the task.



Note. Violin plots depict score density, diamonds indicate the mean and error bars the 95% confidence intervals of the mean.

Discussion

The results from Study 1 showed that increasing the salience of the signal was associated with a significant improvement in Ev. Specifically, the high-salience condition showed a higher proportion of hits and a lower FA rate compared to the low- and standard-

471 salience conditions. These findings support the idea that greater perceptual salience of
472 relevant stimuli, achieved by reducing perceptual task demands, helps maintain a more stable
473 level of vigilance (Helton & Warm, 2008). In contrast, the low- and standard salience
474 conditions replicated the classic deterioration pattern observed in the ANTI-Vea task,
475 characterized by a progressive decline in hit rates. Note, however, that a similar effect over
476 time was observed for the three levels of perceptual salience with sensitivity (A') or response
477 bias (B'') measures.

478 As expected, perceptual salience manipulations did not significantly affect AV
479 measures, as shown in the supplementary material, replicating prior findings (Luna, Barttfeld,
480 et al., 2022). This suggests that, although increased salience of relevant stimuli is effective in
481 supporting goal-directed attention, it does not influence the ability to maintain a general state
482 of alertness to unpredictable temporal signals.

483 Despite the above-mentioned differences in EV indices, perceived mental workload,
484 as indexed by the global NASA-TLX score, did not vary as a function of perceptual salience.
485 Participants reported comparable workload after completing the vigilance task across all
486 salience conditions, indicating that the benefits of increased perceptual salience were not
487 accompanied by subjective reductions in workload. This dissociation between objective
488 performance and subjective workload runs counter to predictions derived from the resource
489 depletion hypothesis and suggests that perceptual facilitation may improve vigilance
490 efficiency without substantially altering perceived task demands.

491 Study 2

492 Building on the results obtained in Study 1 on signal salience, as well as on evidence
493 from a different study showing effects of task load on EV performance (Luna, Barttfeld, et

al., 2022), Study 2 aims to replicate and extend these findings by combining the two dimensions orthogonally in a full within-participant design. Specifically, we examined how different levels of task load (elicited by a single, dual, or triple task, as in Luna, Barttfeld, et al., 2022) and the target's perceptual salience (low, standard, or high, as in Study 1) would modulate the EV decrement over time, as measured via adapted versions of the online ANTI-Vea task. This approach allows for the examination of not only the main effects of task load and the target's salience but also their potential interactions.

We anticipated different predictions according to the resource depletion hypothesis and mind-wandering hypothesis regarding the effects of task load and perceptual salience on the EV decrement. Following the resource depletion hypothesis, the highest overall hit rate and no decrease over time of hits in low demand task conditions was predicted—namely, the single task and high perceptual salience condition—compared to the high demand task conditions—that is, the triple task and the low perceptual salience condition. In contrast, according to the mind-wandering theory, it is expected that a less demanding task could increase task monotony and therefore exacerbate the vigilance decrement. Thus, probably a larger decrement would be observed in the single task load condition and in the high perceptual salience condition, than in more demanding conditions, such as the triple task and the low perceptual salience.

We also expected to replicate outcomes observed in previous studies conducted in our lab. Following Luna, Barttfeld, et al. (2022), we expected that task load would modulate the EV decrement, at least in the standard salience condition. In particular, we expected to observe a similar decrement in the single- and triple-task conditions, but no significant change across time-on-task in hit rate in the dual-task condition, replicating the pattern

reported by Luna, Barttfeld, et al. (2022). Additionally, following the outcomes observed in Study 1, we hypothesized that perceptual salience would modulate the vigilance decrement specifically in the triple task condition, showing no significant change in hit rate in the high salience condition but pronounced decrements in the standard and low salience conditions. We also expected a higher FA rate in the low compared to the standard and high perceptual salience conditions in the triple task load condition, as reported in Study 1.

This study was pre-registered on OSF (<https://osf.io/csab3/wiki/home/>). Note that, however, hypotheses related to self-reported mind-wandering and cognitive control measured via the ANTI-Vea in the triple task condition, were not analyzed here as they exceed the main objectives of the present manuscript and will be reported elsewhere.

Method

Participants

Previous research with between-participants designs conducted with the ANTI-Vea observed that the decrement in hits was modulated by (a) task load, with $n = 24$ participants per group (Luna, Barttfeld et al., 2022), and (b) perceptual salience, with $n = 28$ participants per group (Study 1). In addition, the decrement in hits has been usually observed via the standard ANTI-Vea in studies with $n = 40$ participants per group (Luna et al., 2018; Luna, Barttfeld et al., 2021).

Considering all factors, and noting that in the present study task load and perceptual salience were manipulated within-participant across nine experimental sessions, we decided to gather data from 50 participants. We estimated that this sample size would be sufficient to replicate previous modulations of the vigilance decrement by task load (Luna, Barttfeld et al.,

2022) and perceptual salience (Study 1). Data from 25 participants were collected at the University of Granada, Spain, where each participant received €60 for their participation, whereas the remaining 25 participants were from the National University of Córdoba, Argentina, and voluntarily participated in the study without receiving any economic or academic compensation. Inclusion criteria were being between 18 and 35 years old, having normal or corrected-to-normal vision, and having access to a computer with an internet connection. All participants volunteered to take part in this study and completed an online informed consent form at the beginning of the experiment.

The final sample included 45 participants: 21 from the National University of Córdoba (UNC, Argentina) and 24 from the University of Granada (UGR, Spain). At UNC, 4 participants dropped out for unknown reasons. At UGR, one participant did not complete all sessions and was excluded from analysis.

The study at UGR was approved by the Ethics Committee of the University of Granada (1188/CEIH/2020), whereas at UNC the study was approved by the Ethics Committee of the Institute of Psychological Research (CEIIPsi – PE 41 v.2). The study was conducted in accordance with the principles of the Declaration of Helsinki (latest update in Brazil 2013).

Procedure and Design

Participants completed ten online sessions in total. The first one was a practice session completed along with the experimenter in a video-call that aimed to familiarize participants with the task and procedure. During this session, the use of the web platforms was explained and participants completed the standard version of the ANTI-Vea (triple task with standard perceptual salience). This session allowed them to understand the study procedure, ask questions, and

562 become acquainted with the general characteristics of the task. Importantly, data from this ses-
 563 sion were not included in the analyses.

564 Participants were asked to complete the remaining nine experimental sessions in a
 565 controlled environment of their choice (quiet, well-lit, and with a stable internet connection),
 566 with the speaker volume set to 75% and trying to complete the task at the same time of the
 567 day every session (one session per day). Participants were instructed via email on how to
 568 select the task parameters for each session, being unaware of the specific task load and
 569 perceptual salience manipulations that were implemented in each session.

570 The following sessions were conducted on different days, encouraging participants to
 571 complete the next session in an interval between 24 to 72 hours from the last session, if
 572 possible. In each experimental session, participants completed one of the nine orthogonal
 573 combinations of task load (simple, dual, or triple) and perceptual salience (low, standard, or
 574 high), in a randomized order by participant. The experimental blocks lasted 32 minutes and
 575 48 seconds, consisting of six consecutive blocks of 80 trials (48 ANTI, 16 EV, and 16 AV),
 576 presented randomly and without breaks.

577 Before each session, participants were given specific instructions and practice blocks
 578 according to the task type, to complete only the sub-task or sub-tasks of the ANTI-Vea of that
 579 session, as in Luna, Barttfeld et al. (2022). In addition, before and after the experimental task,
 580 participants completed a series of self-report questionnaires, that is, similar to the Study 1, the
 581 Dundee Stress State Questionnaire (pre- and post-task versions), SAM, DES, the NASA-TLX
 582 (Nygren, 1991), and visual analog scales for intentional and spontaneous mind-wandering at
 583 different time points of the task. The Intentional and Spontaneous Mind-Wandering
 584 Questionnaire (Cásedas et al., 2022) was also administered in the first session. Full details of

these measures and their use are reported elsewhere in relation to the objectives of a larger research project.

Due to unknown reasons, 16 participants involuntarily repeated at least one of the nine sessions. To complete the nine sessions of the experimental design, 9 participants from UGR and 7 from UNC completed an additional session and one from UGR completed four additional sessions. Only the data from the second session were retained for analyses.

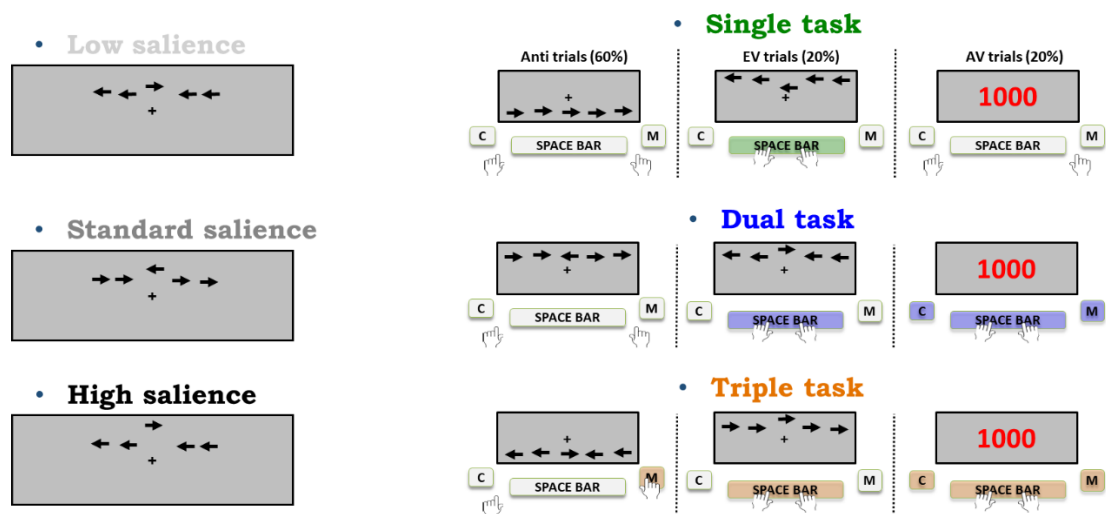
Behavioral Task: ANTI-Vea

The design of the ANTI-Vea task was the same as in Study 1. However, here, an additional manipulation of task load was introduced through task instructions, while keeping the stimuli and their duration constant across all conditions (as depicted in Figure 1A and 1B). Task load was varied at three levels, replicating the manipulation in Luna, Barttfeld et al. (2022). In the single task condition, participants were required to respond only to EV trials by pressing the space bar when the target was vertically displaced (highlighted in green in Figure 4), while not responding when the target was not displaced (i.e., ANTI trials) or to the countdown (i.e., AV trials). In the dual task condition, in addition to responding to EV trials, participants had to stop the countdown by pressing any key in AV trials (highlighted in blue in Figure 4), while not responding when the target was not displaced (i.e., ANTI trials). Finally, in the triple task condition, which is the standard ANTI-Vea task (Coll-Martín et al., 2024; Luna et al., 2018; Luna, Barttfeld et al., 2021), participants had to respond to all three types of trials (highlighted in orange in Figure 4): to the direction pointed by the target in ANTI trials, to detect the vertical displacement of the target in EV trials, and to stop the countdown as fast as possible in AV trials. The orthogonal combination of the three levels of

task load (single, dual, and triple) and perceptual salience (low, standard, and high) resulted in a total of nine distinct experimental conditions.

Figure 4

Representation of perceptual salience (target's displacement) and task load (subtasks to complete) levels manipulated in the procedure of the ANTI-Vea task.



612

Note: The figure illustrates the different levels of perceptual salience (left panel) and task load (right panel). The correct response for each subtask in task load levels is highlighted in green (single task), blue (dual task), and orange (triple task), or not highlighted (i.e., hands off the keyboard) if the task was not intended to be completed.

Mental Workload: NASA-TLX

As in Study 1, perceived mental workload was assessed using the Spanish translation (Arquer & Nogareda, 1999) of the short version of the NASA-TLX (Nygren, 1991). The instrument and scoring procedure were identical to those described in Study 1.

620

621 Data Analysis

622 As in Study 1, to detect any outliers, we checked whether any participant exhibited a
623 false alarm rate exceeding 50%; none met this criterion.

624 To analyze the effects of perceptual salience and task load on the EV decrement,
625 linear mixed-effects models (LME) were used. Analyses were conducted in R 4.2.0 (R Core
626 Team, 2024) with RStudio 2022.02.3 (Posit, 2024), employing the lme4 package (Bates et al.,
627 2015) and lmerTest (Kuznetsova et al., 2017).

628 The models included block (6 levels), perceptual salience (low, standard, high), task
629 load (single, dual, triple), and sample (UGR or UNC) as fixed effects. Random effects for
630 participants and number of sessions were included to account for their variability. Although
631 LME models were not specifically anticipated in our pre-registered protocol, we decided to
632 add random effects given the different number of sessions completed by participants. Note
633 that dependent and independent variables (i.e., fixed effects) were included as anticipated in
634 our pre-registered protocol. LME models were fitted using restricted maximum likelihood
635 and degrees of freedom were estimated using the Satterthwaite method. Effect sizes for post
636 hoc comparisons were calculated using the effectsize package (Ben-Shachar et al., 2020).

637 To assess performance decline over time, hits, FA, sensitivity, and response bias were
638 calculated as in Study 1 and included as dependent variable in separated LME models. Main
639 effects and interactions were examined using ANOVAs on each fitted model.

640 Mental workload, indexed by the NASA -TLX score, was analyzed using a repeated-
641 measures ANOVA, with perceptual salience and task load as within-participant factors. Four
642 participants were excluded from this analysis due to technical issues resulting in incomplete
643 data.

Results

For the sake of clarity, the main effect of sample was not observed as significant for hits, FA, sensitivity, or response bias as dependent variable (all $ps > .05$). Therefore, no significant differences between samples from UGR and UNC were observed, supporting data collapsing.

Table 3 shows average scores for EV measures as a function of the perceptual salience and task load.

Table 3

Overall scores for executive vigilance as a function of the perceptual salience and task load.

Percep- tual sa- lience	Task load	Hits		False alarms		A'		B''	
		M (%)	95% CI	M (%)	95% CI	M	95% CI	M	95% CI
Low	Sin- gle	78.19	[78.16,	13.8	[11.54,	.8	[.882,	.2	[.08, .32]
			78.22]	7	16.20]	9	.898]	0	
	Dual	79.98	[79.95,	17.7	[14.82,	.8	[.870,	.0	[-.09, .13]
			80.01]	8	20.74]	8	.890]	2	
	Tri- ple	70.17	[70.14,	9.67	[7.96,	.8	[.870,	.4	[.35, .51]
			70.20]		11.38]	8	.890]	3	
Standard	Sin- gle	87.70	[87.68,	7.99	[6.19, 9.79]	.9	[.931,	.0	[-.07, .15]
			87.72]			4	.949]	4	
	Dual	89.95	[89.93,	8.77	[6.89,	.9	[.933,	-	[-.15, .11]
			89.97]		10.65]	4	.947]	.0	

	Tri-	82.29	[82.27,	4.35	[2.90, 5.80]	.9	4	[.932,	.4	[.33, .55]
	ple		82.31]					.948]	4	
High	Sin-	93.33	[93.31,	3.21	[1.79, 4.63]	.9	7	[.961,	-	[-.26, .00]
	gle		93.35]					.979]	.1	
									3	
	Dual	94.49	[94.48,	3.44	[2.17, 4.71]	.9	7	[.964,	-	[-.45, -
			94.50]					.976]	.3	.19]
									2	
	Tri-	91.20	[91.18,	2.12	[0.63, 3.61]	.9	7	[.964,	.1	[-.02, .30]
	ple		91.22]					.976]	4	

653

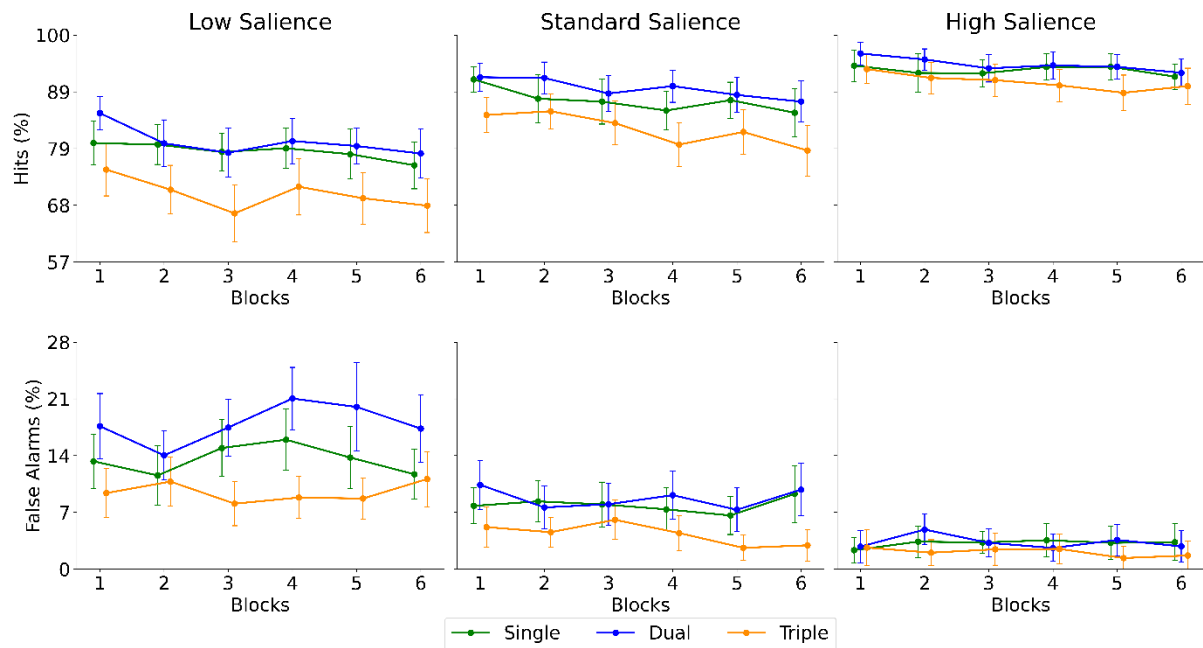
654 *Note.* M = mean, 95% CI = 95% CI of the mean were computed following the Cousineau-
655 Morey method (Morey, 2008).

656 Hits

657 A significant main effect of perceptual salience on hits was found (see Table 4). The
658 largest overall hit rate was observed in the high salience condition, which was significantly
659 different than that in the standard salience condition, $t(2329.38) = 10.16, p < .001, d = 2.26$
660 $[1.70, 2.81]$, and in the low salience condition, $t(2329.82) = 26.66, p < .001, d = 5.92 [4.91,$
661 $6.93]$. The hit rate was significantly larger in the standard than in the low salience condition,
662 $t(2326.21) = 16.93, p < .001, d = 3.76 [3.03, 4.48]$ (see Table 3).

663 Figure 5

664 *Hits and false alarms across time-on-task as a function of the perceptual salience and task*
665 *load.*



666

667 *Note.* Error bars represent 95% CI of the mean and were computed following the Cousineau-
 668 Morey method (Morey, 2008).

669 **Table 4**670 *Repeated-measures ANOVAs for executive vigilance measures.*

DV	Main effect or interaction	<i>F</i>	df	<i>p</i>	η^2_p [95% CI]
Hits	Perceptual Salience	365.2	2, 2325.37	< .001	.24 [.21, .27]
	Task Load	73.33	2, 2327.86	< .001	.06 [.04, .08]
	Perceptual Salience * Task Load	7.70	4, 2324.23	< .001	.00 [.00, .00]
	Block	8.13	5, 2307.99	< .001	.02 [.01, .03]

	Perceptual Salience * Block	0.87	10, 2307.99	.552	.00 [.00, .01]
	Task Load* Block	0.29	10, 2307.99	.982	.00 [.00, .00]
	Perceptual Salience * Task Load*	0.49	20, 2307.99	.97	.00 [.00, .00]
	Block				
	Perceptual Salience	296.0	2, 2329.41	<	.20 [.17, .23]
		2		.001	
	Task Load	62.44	2, 2326.26	<	.05 [.03, .07]
				.001	
	Perceptual Salience * Task Load	13.07	4, 2329.53	<	.02 [.01, .03]
				.001	
False	Block	0.59	5, 2320.27	.705	< .01 [.00,
Alarms					.00]
	Perceptual Salience * Block	1.19	10, 2320.27	.292	< .01 [.00,
					.01]
	Task Load* Block	0.62	10, 2320.27	.798	< .01 [.00,
					.00]
	Perceptual Salience * Task Load*	1.13	20, 2320.27	.307	.01 [.00, .01]
	Block				
	Perceptual Salience	659.6	2, 2332	<	.36 [.33, .39]
		8		.001	
Sensiti-	Task Load	2.33	2, 2332	.097	< .01 [.00,
vity (A')					.01]
	Perceptual Salience * Task Load	2.16	4, 2332	.071	.004 [.00,

				.01]
Block	5.19	5, 2332	<	.01 [.00, .02]
				.001
Perceptual Salience * Block	1.14	10, 2332	.328	< .01 [.00, .01]
Task Load* Block	0.10	10, 2332	.9998	< .01 [.00, .00]
Perceptual Salience * Task Load*	0.57	20, 2332	.932	< .01 [.00, .01]
Block				
Perceptual Salience	45.42	2, 2283.03	<	.04 [.02, .05]
				.001
Task Load	85.04	2, 2264.72	<	.07 [.05, .09]
				.001
Perceptual Salience * Task Load	1.20	4, 2298.27	.310	< .01 [.00, .01]
Response				
bias (B'')				
Block	4.04	5, 2321.73	.001	.01 [.00, .02]
Task Load* Block	0.80	10, 2321.73	.633	< .01 [.00, .00]
Perceptual Salience * Block	1.05	10, 2321.73	.402	< .01 [.00, .01]
Perceptual Salience * Task Load*	0.62	20, 2321.73	0.90	< .01 [.00, .00]
Block				

671 *Note.* Significant outcomes are highlighted in bold. Abbreviations: df, degrees of freedom,
 672 DV, Dependent Variable.

673 As shown in Table 4, a significant main effect of task load was observed. Although
 674 the hit rate was highest in the dual-task condition, the difference between the dual and single-
 675 task conditions, while statistically significant ($t(2331.58) = 2.42, p = .040$), was very small in
 676 magnitude ($d = 0.10 [0.02, 0.18]$). In contrast, the comparisons involving the triple-task
 677 condition showed larger effects: dual vs. triple ($t(2326.08) = 11.50, p < .001, d = 0.48 [0.39,$
 678 $0.56]$) and single vs. triple ($t(2331.96) = 9.06, p < .001, d = 0.38 [0.29, 0.46]$). These results
 679 suggest that the most meaningful changes in hit rates occurred when task demands increased
 680 from dual to triple load (see Table 3).

681 We also observed a significant interaction between perceptual salience and task load
 682 (see Table 3). In the low salience condition, hit rate was significantly larger in the single task,
 683 $t(2330.58) = 8.38, p < .001, d = 0.35 [0.27, 0.43]$, and the dual task, $t(2320.46) = 9.99, p$
 684 $< .001, d = 0.41 [0.33, 0.50]$, compared to the triple task (see Table 3). A similar pattern was
 685 observed in the standard salience condition, with a larger hit rate in the single task, $t(2321.08)$
 686 $= 5.65, p < .001, d = 0.23 [0.15, 0.32]$, and in the dual task, $t(2331.01) = 7.08, p < .001, d =$
 687 $0.29 [0.21, 0.37]$, than in the triple task. *Instead, in the high salience condition, hit rates were*
 688 *significantly larger in the dual task compared to the triple task, $t(2321.17) = 2.95, p = .0032,$*
 689 *$d = 0.12 [0.04, 0.20]$. The difference between the single and triple-task conditions did not*
 690 *reach statistical significance ($t(2333.16) = 1.82, p = .0695$). In all perceptual salience levels,*
 691 *no significant differences were found between the single and the dual task (all p 's $> .05$).*

692 As in Study 1, and consistent with previous findings using the ANTI-Vea task (Coll-
 693 Martín et al., 2023; Luna et al., 2018; Luna, Roca, et al., 2021), a significant change in hit

rate across blocks was observed (see Table 4), with a significant and negative linear component, $t(2319.25) = -5.82, p < .001, \eta_p^2 = .01, [.01, .03]$.

As shown in Figure 5 and Table 4, the change in hits across blocks was not significantly modulated by perceptual salience, nor by task load. Finally, the three-way interaction among task load, perceptual salience, and block was not significant.

In line with our preregistered hypotheses and aiming to replicate previous findings from our laboratory (Luna, Barttfeld, et al., 2022), we conducted planned simple-effects analyses with partial ANOVAs, focusing on specific levels of perceptual salience and task load. Specifically, task load was examined under standard perceptual salience conditions, and perceptual salience was examined within the triple-task condition. These analyses were hypothesis-driven and preregistered, rather than exploratory follow-ups to a significant interaction. Task load did not modulate hits across blocks when considering only the standard salience condition, $F(10, 739.71) = 0.53, p = .862, \eta_p^2 = .00, [.00, .01]$, in contrast to the results reported in Luna, Barttfeld, et al. (2022). Finally, perceptual salience did not modulate hit rate across blocks when considering only the triple task condition, $F(10, 736.63) = 1.09, p = .362, \eta_p^2 = .01, [.00, .02]$, as observed in Study 1.

False alarms

As reported in Table 4, a significant main effect of perceptual salience on FA was observed. Similarly to the pattern observed in Study 1, the largest FA rate was observed in the low salience condition, which was significantly different than that in the standard salience condition, $t(2329.02) = 14.98, p < .001, d = 0.62 [0.54, 0.70]$, and in the high salience condition, $t(2320.35) = 24.04, p < .001, d = 1 [0.91, 1.08]$. FA rate was significantly larger in

716 the standard than in the high salience condition, $t(2326.78) = 9.40, p < .001, d = 0.39 [0.31,$
 717 $0.47]$.

718 The main effect of task load was also significant (see Table 4). The smallest FA rate
 719 was observed in the triple task condition, which was significantly different than that in the
 720 single task condition, $t(2325.56) = -7.03, p < .001, d = -0.29 [-0.37, -0.21]$, and in the dual
 721 task condition, $t(2331.84) = -11.03, p < .001, d = -0.46 [-0.54, -0.37]$. In addition, FA rate
 722 was significantly smaller in the single than in the dual task condition, $t(2322.87) = -4.01, p$
 723 $= .0002, d = -0.17 [-0.25, -0.09]$.

724 As observed for hit rate, a significant interaction between task load and perceptual
 725 salience was observed on FA rate (see Table 4). In the low salience condition, the smallest FA
 726 rate was observed in the triple task, which was significantly different than that in the dual,
 727 $t(2329.24) = 11.45, p < .001, d = 0.47 [0.39, 0.55]$, and in the single task, $t(2331.91) = 5.66, p$
 728 $< .001, d = 0.23 [0.15, 0.32]$. Moreover, FA rate was significantly larger in the dual than in
 729 the single task, $t(2331.73) = 5.83, p < .001, d = 0.24 [0.16, 0.32]$. In the standard salience
 730 condition, the triple task again showed the smallest FA rate, which was significantly different
 731 than that in the single, $t(2331.32) = 5.22, p < .001, d = 0.22 [0.13, 0.30]$, and in the dual task,
 732 $t(2327.98) = 6.25, p < .001, d = 0.26 [0.18, 0.34]$. However, FA rate was not significantly
 733 different between the single and the dual task, $t(2328.51) = 1.07, p = .285, d = 0.04 [0.04,$
 734 $0.13]$. Interestingly, in the high salience condition, there were no significant differences on FA
 735 rate between task load conditions; triple vs. single task, $t(2305.50) = 1.39, p = .166, d = 0.06$
 736 $[-0.02, 0.14]$, triple vs. dual task, $t(2331.66) = 1.51, p = .131, d = 0.06 [-0.02, 0.14]$, and
 737 single vs. dual task, $t(2319.59) = 0.11, p = .910, d = 0.00 [-0.08, 0.09]$.

The main effect of block was not observed as significant for FA rate (see Figure 5).

Lastly, as observed for hit rate, no significant interactions were observed between block and perceptual salience, block and task load, and between task load, perceptual salience, and block (see Figure 5 and Table 3).

To test the pre-registered hypothesis, a partial ANOVA was conducted to examine whether perceptual salience modulated FA rates within the triple task condition. A significant main effect of perceptual salience was observed, $F(2, 745.54) = 74.60, p < .001, \eta_p^2 = .17, [0.12, 0.21]$. The largest FA rates were observed in the low salience condition, which were significantly different than the standard salience condition, $t(749.29) = 8.56, p < .001, d = 1.90 [1.37, 2.42]$, and in the high salience condition, $t(731.90) = 11.79, p < .001, d = 2.62 [2.02, 3.21]$. FA rates were significantly larger in the standard salience condition than in the high salience condition, $t(749.82) = 3.50, p = .0014, d = 0.78 [0.32, 1.23]$.

Sensitivity

A significant main effect of perceptual salience on A' was found (see Table 3). Overall A' was significantly larger in the high than in the low salience condition, $t(1663.63) = 35.37, p < .001, d = 1.73 [1.62, 1.85]$; in the high than in the standard salience condition, $t(1780.69) = 11.91, p < .001, d = 0.56 [0.47, 0.66]$; and in the standard than in the low salience condition, $t(2325.03) = 23.65, p < .001, d = 0.98 [0.89, 1.07]$.

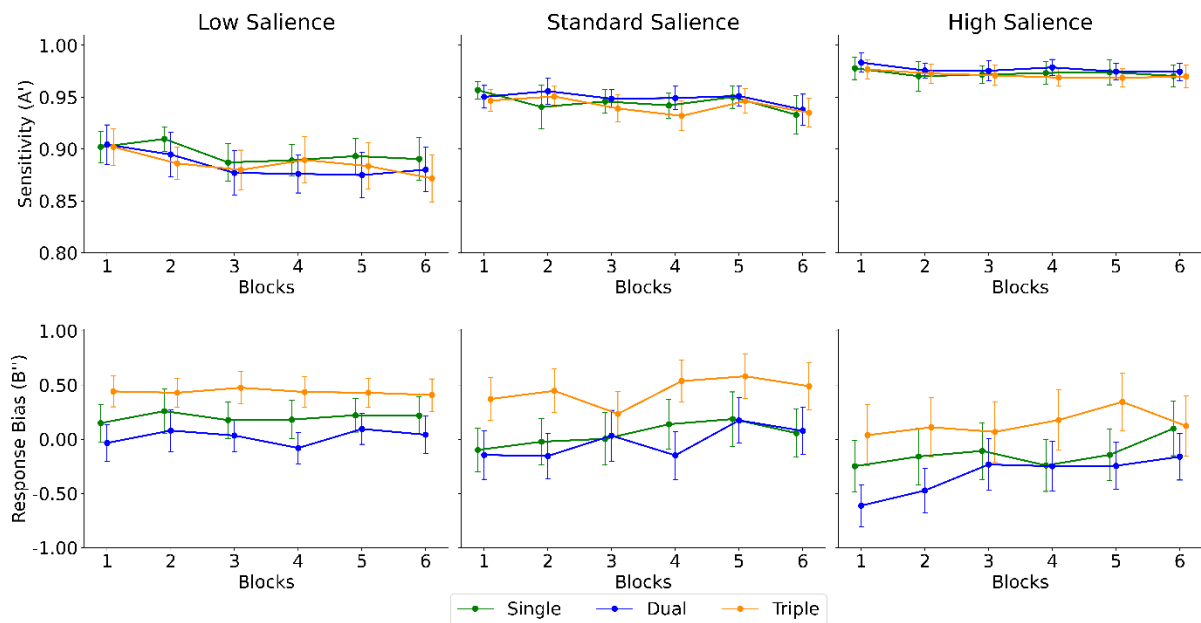
In contrast, as shown in Table 3, the main effect of task load and the interaction between task load and perceptual salience were not observed as significant.

A significant main effect of block was observed (see Table 3), with a significant and negative linear component, $t(2322.81) = -4.45, p < .001, \eta_p^2 = .00, [.00, .02]$, showing therefore a decrease in A' across blocks.

As can be seen in Table 3, no significant interactions were found between block and perceptual salience, between block and task load, or between task load, perceptual salience, and block (see Figure 6).

Figure 6

Sensitivity and Response Bias across time-on-task as a function of the perceptual salience and task version.



Note. Error bars represent 95% CI of the mean and were computed following the Cousineau-Morey method (Morey, 2008).

Response bias

A significant main effect of perceptual salience on B'' was observed (see Table 3). Overall B'' was significantly larger in the low than in the high salience condition, $t(2206.40) = 9.02, p < .001, d = 0.38 [0.30, 0.47]$; in the standard than in the high salience condition, $t(2249.19) = 7.17, p < .001, d = 0.30 [0.22, 0.39]$, but not significantly different between the low and the standard salience condition, $t(2331.48) = 1.91, p = .137, d = 0.08 [0.00, 0.16]$.

Also, a significant main effect of task load was found. Overall B'' was significantly larger in the triple task than in the dual task, $t(2262.00) = 12.78, p < .001, d = 0.54$ [0.45, 0.62]; in the triple task than in the single task, $t(2268.96) = 8.49, p < .001, d = 0.36$ [0.27, 0.44]; and in the single task than in the dual task, $t(2265.72) = 4.34, p < .001, d = 0.18$ [0.10, 0.26]. As shown in Table 3, the interaction between task load and perceptual salience was not significant.

The main effect of block was observed as significant, with a significant and positive linear component, $t(2322.81) = 4.14, p < .001, \eta_p^2 = .00, [.00, .02]$, showing therefore a change in response bias towards a more conservative criterion. Lastly, as shown in Figure 6, interactions between block and task load, block and perceptual salience, and between block, perceptual salience, and task load were not significant.

Mental Workload: NASA-TLX

For the global NASA-TLX score, unlike the result of Study 1, there was a significant main effect of perceptual salience, $F(1.95, 78.03) = 3.25, p = .045, \eta_p^2 = .08$ [.00, .19].

Overall, NASA-TLX scores were significantly higher in the low salience than in the high salience condition, $t(40) = 2.69, p = .028, d = 0.85$ [0.20, 1.49] (see Figure 7). In contrast, workload did not differ significantly between the standard and high salience conditions, $t(40) = 0.92, p = .633, d = 0.29$ [-0.33, 0.91], nor between the low and standard salience conditions, $t(40) = 1.50, p = .301, d = 0.47$ [-0.16, 1.10].

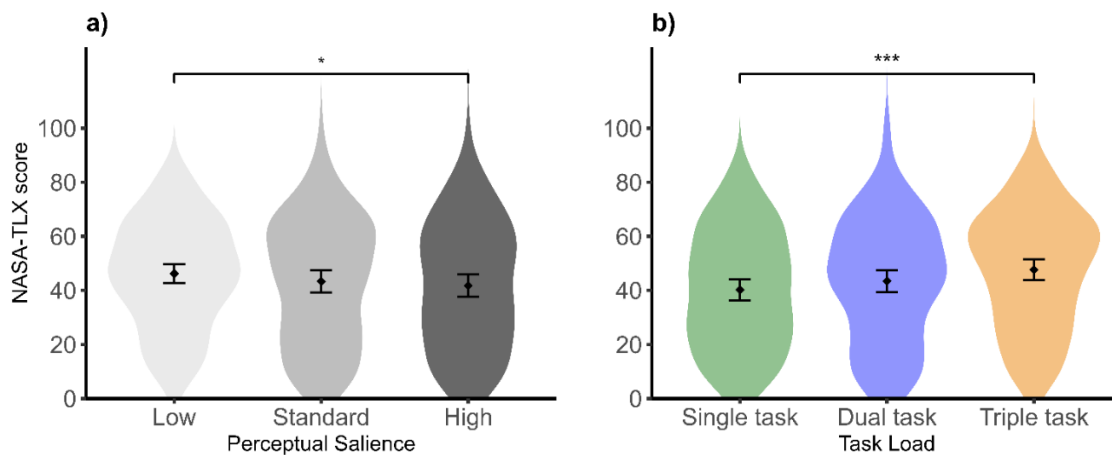
In addition, a significant main effect of task load was observed, $F(1.78, 71.19) = 8.93, p < .001, \eta_p^2 = .18$ [.05, .32]. NASA-TLX scores were significantly higher in the triple task than in the single task condition, $t(40) = 4.32, p < .001, d = 1.37$ [0.67, 2.05]. In contrast, workload did not differ significantly between the triple task and the dual task, $t(40) = 2.06, p$

799 = .111, $d = 0.65$ [0.01, 1.28], nor between the single task and dual task, $t(40) = -2.17$, p
 800 = .090, $d = -0.68$ [-1.32, -0.04].

801 The interaction between perceptual salience and task load was not significant, $F(3.46,$
 802 $138.27) = 0.82$, $p = .497$, $\eta_p^2 = .02$.

803 **Figure 7**

804 *Mental workload (mean NASA-TLX score) as a function of the perceptual salience and task*
 805 *load of the task.*



806
 807 *Note.* Violin plots depict score density, diamonds indicate the mean, and error bars the 95%
 808 confidence intervals of the mean. * $p < .05$, *** $p < .001$

809 **Discussion**

810 The results of Study 2 revealed clear main effects of both experimental manipulations,
 811 perceptual salience and task load. Consistent with the findings from Study 1, perceptual
 812 salience modulated overall hits and false alarms: participants detected more targets and made
 813 fewer false alarms when the target stimuli in the high salience condition. Regarding the pre-
 814 registered hypothesis that FA rates would be higher in the low salience condition than in
 815 standard and high salience ones, the current outcomes replicated those of Study 1. These

results could support the predictions of the resource depletion hypothesis, which posits that less salient stimuli increase task demands. This, in turn, leads to greater consumption of attentional resources and consequently poorer performance (Warm et al., 2008; Helton & Warm, 2008). This interpretation, however, is not supported by the results observed for sensitivity (A'), as no corresponding effect was observed. Importantly, no modulation of perceptual salience over the EV decrement was observed with any dependent variable.

Subjective workload measures converged with these performance effects. Specifically, NASA-TLX scores were higher in the low salience condition compared to the high salience condition, indicating that reduced perceptual salience was experienced as more mentally demanding. This pattern supports the interpretation that low salience increases perceived effort, even in the absence of changes in sensitivity.

On the other hand, the manipulation of task load showed a non-linear pattern on hits: the dual-task condition led to better hit rates than both the single and triple-task conditions, suggesting a benefit of maintaining a moderate level of mental workload. This interpretation is strongly supported by the NASA-TLX results, which showed substantially higher workload in the triple-task condition compared to the single-task condition, indicating that the task load manipulation effectively altered perceived mental demand. The lower hit rates observed in the single-task condition are consistent with the mind-wandering hypothesis, which proposes that low task demands facilitate attentional disengagement and mind-wandering (Seli et al., 2018). In contrast, the decline in hit rates in the triple-task condition is more consistent with the resources depletion account, according to which excessive task demands may exceed available cognitive resources. Taken together, these findings suggest that the relationship between task load and vigilance decrement is better characterized by an inverted-U-shape

pattern, whereby both insufficient and excessive workload impair hit rates, rather than by a single underload-based explanation (Luna, Barttfeld et al., 2022).

Importantly, this inverted-U-shape pattern was observed at the level of hits, while no corresponding differences were found in A' . This suggests that the task load manipulation could influence response bias rather than sensitivity. Recent work on vigilance tasks has highlighted that, particularly in modern paradigms characterized by very low FA, performance decrements may predominantly reflect shifts in response bias rather than changes in sensitivity (Thomson et al., 2016). From this perspective, moderate levels of task demand may promote a more optimal response criterion—supporting sustained engagement—whereas both low and high workload may bias participants toward a more conservative response criterion, leading to reduced hit rates.

This view is supported by previous research indicating that both active fatigue (associated with high load) and passive fatigue (associated with low stimulation) can impair performance in different ways, depending on how individuals regulate mental effort (Saxby et al., 2013). Similarly, a study in semi-automated driving contexts has shown that vigilance decrements emerge when mental workload is too low, highlighting the risks of sustained attention failures under underload conditions (McWilliams & Ward, 2021). Therefore, maintaining an optimal level of cognitive demand may be critical for sustaining attention, as suggested by both theoretical models and empirical findings. Nevertheless, and critically, despite these main effects of task load, no modulation over the EV decrement was observed.

Additionally, a significant interaction between perceptual salience and task load was observed: under low-salience conditions, lower-load tasks (single and dual) supported better performance in hits and FA compared to the triple-task condition. However, this effect

diminished as salience increased, suggesting that more efficient perceptual processing may partially compensate for the detrimental effects of high task load. This interaction highlights the importance of considering multiple sources of task demands when studying vigilance decrement.

Importantly, despite these effects on overall performance, no significant interactions were found between time-on-task and the task load and perceptual salience manipulations. This suggests that while these variables affect the overall level of performance, they do not substantially change the temporal trajectory of vigilance decrement.

Taken together, these findings underscore the need for more integrative models capable of explaining how task load and perceptual salience interact to influence both performance levels and attentional stability during prolonged tasks. They also highlight the utility of the ANTI-Vea paradigm as a sensitive tool for disentangling these effects and assessing the distinct contributions of different components of vigilance.

Study 3: Re-analysis of pooled data

In light of the relevant yet somewhat contradictory findings of Studies 1 and 2, and those observed in previous laboratory-based research (Luna, Barttfeld, et al., 2022) regarding the modulation of perceptual salience and task load on EV, we re-analyzed data collected across different samples and environments (i.e., lab and online) to gain a better and more solid understanding. Thus, the main goal of this reanalysis was to assess the robustness of the perceptual salience and task load manipulations, and more importantly, their modulation over time-on-task on EV measured under the same procedure (i.e., the ANTI-Vea task) across the different samples. [Analyses were focused on signal detection theory measures, that is,](#)

sensitivity (A') and response bias (B''), whereas hit and FA rates are reported in the Supplementary Material.

Whereas the re-analysis of pooled data from a large sample size increases the statistical power of observed outcomes, note that participants' assignment to experimental conditions was not randomized in the present re-analysis. Therefore, the observed outcomes should not be considered as confirmatory, but rather exploratory. We expect that this re-analysis can provide new empirical evidence concerning a more integrative understanding of how EV is influenced by different task demands (i.e., perceptual salience and task load) that can be manipulated either independently or simultaneously in between or within-participant designs.

Method

Participants

The total sample for the re-analysis consisted of 238 participants: 84 from Study 1, 45 from Study 2, and 109 from the study by Luna, Barttfeld et al. (2022). For details, see Table 5.

Table 5

Sample characteristics across studies.

Study characteristics	Study 1	Study 2	Luna, Barttfeld et al. (2022)
Design	Between-participants	Within-participant	Between-participants

Setting (lab/online)		Online	Online	Lab
<i>N</i>		84	45	109
Low salience	Single task		<i>n</i> = 45	
	Dual task		<i>n</i> = 45	
	Triple task	<i>n</i> = 28	<i>n</i> = 45	
Standard salience	Single task		<i>n</i> = 45	<i>n</i> = 45
	Dual task		<i>n</i> = 45	<i>n</i> = 24
	Triple task	<i>n</i> = 30	<i>n</i> = 45	<i>n</i> = 40
High salience	Single task		<i>n</i> = 45	
	Dual task		<i>n</i> = 45	
	Triple task	<i>n</i> = 26	<i>n</i> = 45	

901 Data analysis

902 The pooled data from the different studies were re-analyzed using separate LME for
903 sensitivity (A') and response bias (B''). Each model included block (six levels), task load
904 (single, dual, triple), and perceptual salience (low, standard, high) as fixed effects, together
905 with all second- and third-way interactions. Noting that participants performed either one or
906 several combinations of task demands and completed the procedure either at the lab or online,
907 we decided to add random effects for participants, number of sessions that each participant
908 completed, and sampling procedure, to therefore account for variability associated with these
909 sources. Models were fitted using restricted maximum likelihood and degrees of freedom
910 were estimated using the Satterthwaite method. Main effects and interactions were examined
911 through ANOVAs performed on the fitted LME-model.

Results

Sensitivity

A LME model was fitted with A' as dependent variable (see Table 6). The main effect of perceptual salience was significant. The largest overall A' was observed in the high-salience condition (see Figure 8), which was significantly larger than that in the standard-salience condition, $t(3472.95) = 14.03, p < .001, d = 0.48, [0.41, 0.54]$, and in the low-salience condition, $t(3510.05) = 40.16, p < .001, d = 1.36, [1.28, 1.43]$. A' was also significantly larger in the standard-salience condition than in the low-salience condition, $t(3473.76) = 26.29, p < .001, d = 0.89, [0.82, 0.96]$. The main effect of task load was not significant. The interaction between perceptual salience and task load was significant but very small (see Table 6). As shown in Figure 8, A' increased with perceptual salience across all task-load conditions, with only slight variations in the magnitude of this effect across conditions.

The main effect of block was significant, indicating that A' changed across time-on-task, with a significant and negative linear component, $t(3302.93) = -5.31, p < .001, \eta_p^2 = .008, [.00, .02]$. However, neither the interaction between block and task load nor the interaction between block and perceptual salience were significant. The three-way interaction among block, task load, and perceptual salience was also not significant. Thus, although A' varied as a function of perceptual salience and showed a small interaction between perceptual salience and task load, neither manipulation significantly modulated changes in sensitivity across blocks.

Table 6

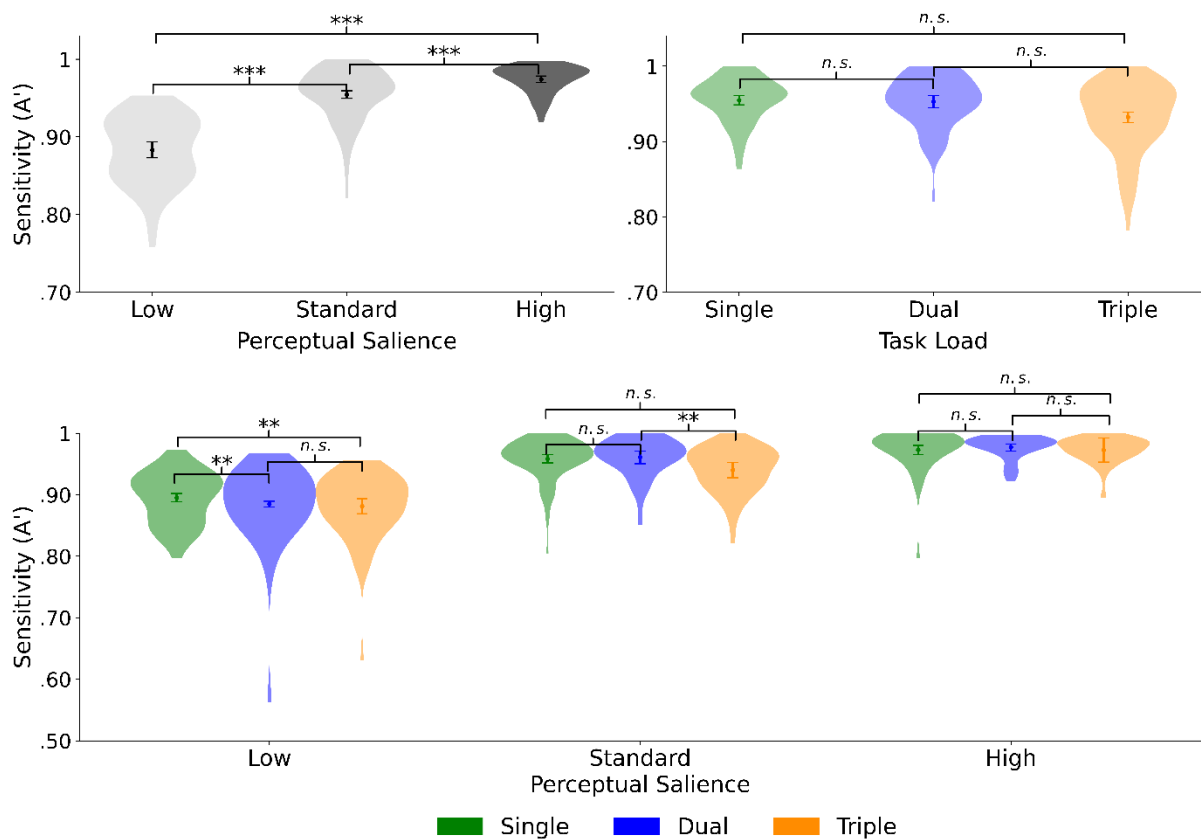
934 *ANOVA on Linear Mixed Effect model for Sensitivity*

Main effect or interaction	<i>F</i>	df	<i>p</i>	η^2_p [95% CI]
Perceptual Saliency	831.75	2, 3493.6	< .001	.32 [.30, .35]
Task Load	2.97	2, 3269.8	.052	.00 [.00, .01]
Task Load * Perceptual Saliency	3.35	4, 3409.4	.010	.00 [.00, .01]
Block	6.75	5, 3302.9	< .001	.01 [.00, .02]
Block * Task Load	0.07	10, 3302.9	.999	.00 [.00, .00]
Block * Perceptual Saliency	1.50	10, 3302.9	.133	.00 [.00, .01]
Block * Task Load * Perceptual Saliency	0.52	20, 3302.9	.961	.00 [.00, .00]

935 *Note.* Significant outcomes are highlighted in bold. Abbreviations: df, degrees of freedom.

936 **Figure 8**

937 *Overall effects of Sensitivity (*A'*) by perceptual saliency and task load.*



Note. * $p < .05$, ** $p < .01$, *** $p < .001$, n.s. = not significant

Response bias

A separated LME model was fitted with B'' as dependent variable (see Table 7). The main effects of perceptual salience and task load were both significant (see figure 9).

Regarding perceptual salience, the lowest overall B'' was observed in the high-salience condition. B'' was significantly lower in the high-salience condition than in the low-salience condition, $t(3193.36) = 9.12$, $p < .001$, $d = 0.32$, $[0.25, 0.39]$, and in the standard-salience condition, $t(2934.58) = 7.58$, $p < .001$, $d = 0.28$, $[0.21, 0.35]$. In contrast, B'' did not significantly differ between the low- and standard-salience conditions, $t(3208.79) = 1.62$, $p = .237$, $d = 0.06$, $[-0.01, 0.13]$. Thus, participants adopted a less conservative response criterion in the high-salience condition than in the low- and standard-salience conditions.

Regarding task load, the largest overall B'' was observed in the triple-task condition, which was significantly higher than that in the single-task condition, $t(2280.33) = 8.51, p < .001, d = 0.36, [0.27, 0.44]$, and in the dual-task condition, $t(2569.16) = 13.45, p < .001, d = 0.53, [0.45, 0.61]$. B'' was also significantly larger in the single-task condition than in the dual-task condition, $t(3353.45) = 5.08, p < .001, d = 0.18, [0.11, 0.24]$. Thus, participants adopted the most conservative response criterion in the triple-task condition, an intermediate criterion in the single-task condition, and the most liberal criterion in the dual-task condition. In this case, the interaction between perceptual salience and task load was not significant.

The main effect of block was significant, indicating that response bias changed across time-on-task, with a significant positive linear trend, $t(3278.75) = 5.71, p < .001, \eta_p^2 = .01, [.00, .02]$. However, neither the interaction between block and task load nor the interaction between block and perceptual salience were significant. The three-way interaction between block, task load, and perceptual salience was also not significant. Therefore, although B'' differed as a function of both perceptual salience and task load, neither factor significantly modulated its change across blocks.

Table 7

ANOVA on Linear Mixed Effect model for Response bias.

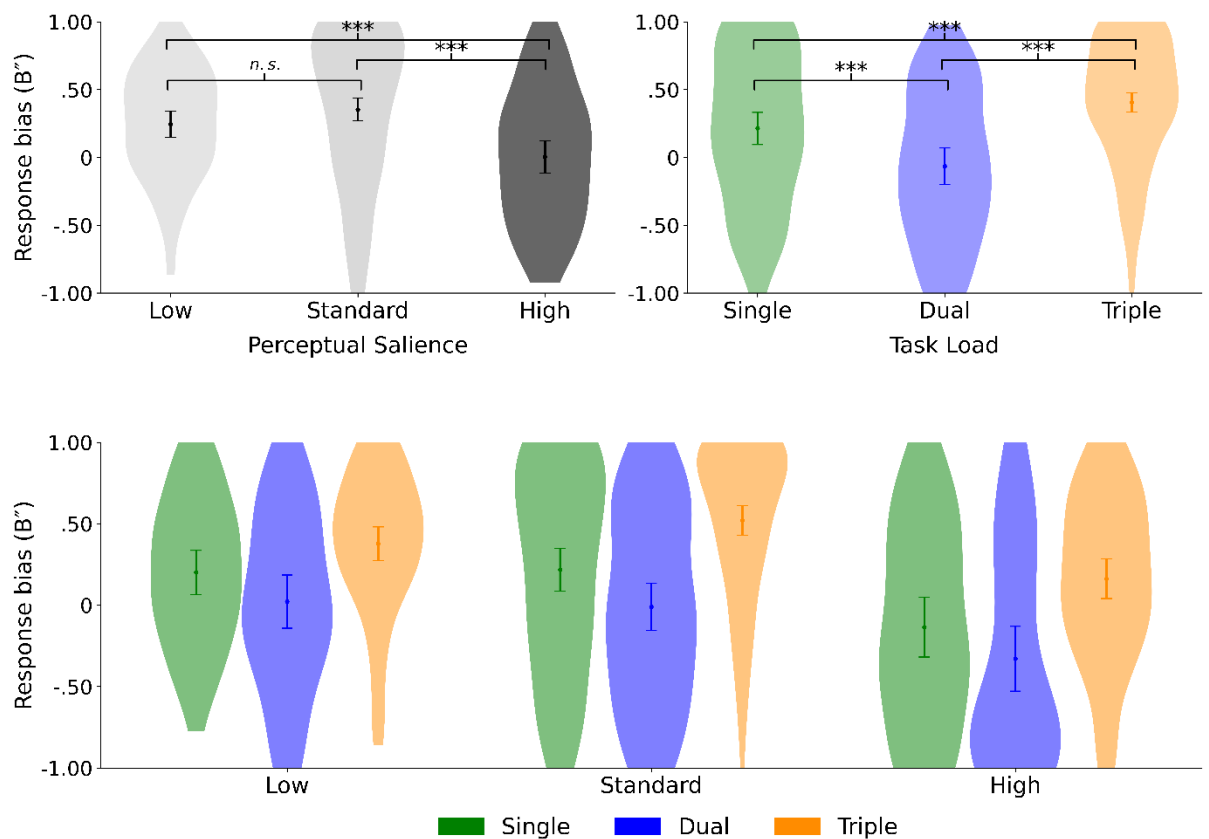
Main effect or interaction	F	df	p	η_p^2 [95% CI]
Perceptual Salience	47.22	2, 3158.7	< .001	.03 [.02, .04]
Task Load	92.11	2, 2533.8	< .001	.07 [.05, .09]
Task Load * Perceptual Salience	1.17	4, 3051.3	.323	.00 [.00, .00]
Block	7.42	5, 3278.8	< .001	.01 [.00, .02]

Block * Task Load	0.24	10, 3278.8	.993	.00 [.00, .00]
Block * Perceptual Salience	0.98	10, 3278.8	.461	.00 [.00, .00]
Block * Task Load * Perceptual Salience	0.69	20, 3278.8	.837	.00 [.00, .00]

Note. Significant outcomes are highlighted in bold. Abbreviations: df, degrees of freedom.

Figure 9

Overall effects of response bias by perceptual salience and task load.



Note. * $p < .05$, ** $p < .01$, *** $p < .001$, n.s. = not significant

General Discussion

The present work analyzed how perceptual salience and task load independently and jointly influence the EV decrement, using the ANTI-Vea task across three complementary

data sets. Study 1 manipulated perceptual salience; Study 2 implemented a full factorial design with both salience and task load; and Study 3, a final re-analysis, replicated this design in a larger more diverse sample ($n = 238$) by combining both studies with data from previous work (Luna, Barttfeld et al., 2022). Importantly, and in contrast to previous research, both target salience and task load were manipulated while maintaining all other aspects of the task, stimuli, and temporal parameters constant across conditions.

Across all three studies, the overall result showed that increasing perceptual salience consistently enhanced performance: higher salience levels were associated with higher hit rates, fewer FA, and greater sensitivity. In Study 1, the high-salience condition was associated with a higher proportion of hits and a lower FA rate than the low- and standard-salience conditions. Study 2 replicated and extended this pattern in a within-participant design, showing that salience affected EV even when combined with different levels of task load. Finally, Study 3 confirmed the robustness of this effect in the pooled sample.

Critically, analyses on signal detection theory measures clarified the nature of the salience effect. Higher perceptual salience was associated with largest sensitivity, indicating that participants were better able to discriminate signal from non-signal trials when the target displacement was more salient. This graded pattern was especially clear in Study 3, where sensitivity was largest in the high-salience condition, followed by the standard- and low-salience conditions. Therefore, the salience manipulation was effective not only in changing direct performance measures, such as hits and FA, but also in modifying the discriminability of the EV signal.

This pattern is generally consistent with the idea that increasing the perceptual salience of critical signals reduces perceptual task demands and facilitates target detection

(Helton & Warm, 2008; Warm et al., 2008). However, the findings also suggest that this interpretation should be qualified. In Study 1, although high salience improved overall performance, sensitivity did not show the same time-on-task modulation observed in hits. Moreover, the NASA-TLX results in Study 1 indicated that subjective workload did not significantly differ as a function of perceptual salience. This suggests that increasing signal salience may improve perceptual discrimination without necessarily producing a corresponding reduction in perceived workload.

Task load, in contrast, showed a non-linear effect. In line with the findings of Luna et al. (2022), participants performed better under dual-task conditions compared to both single and triple-task conditions. This inverted-U pattern aligns with mind-wandering hypothesis and resource control theory (Seli et al., 2018; Thomson et al., 2015), which suggest that moderate task complexity promotes sustained attentional engagement. Performance declined when tasks were either too simple (low stimulation) or too demanding (overload), likely due to disengagement or fatigue. Moreover, a significant interaction between salience and task load was observed in Study 2 and the final summary reanalysis (Study 3): under low-salience conditions, performance was particularly impaired by high task load, while high salience mitigated this negative effect. This underscores the importance of jointly considering perceptual and task load when designing vigilance tasks.

However, the signal detection analyses indicate that this interpretation should be substantially softened. Task load did not produce a robust general effect on sensitivity. In contrast, it clearly affected response bias. Participants adopted the most conservative criterion in the triple-task condition, an intermediate criterion in the single-task condition, and the most liberal criterion in the dual-task condition. Therefore, the differences observed in hit and

FA rates across task-load conditions should not be interpreted as direct evidence of differences in signal discrimination. Rather, they seem to reflect, at least in part, changes in response strategy. The lower hit rates observed in the single-task condition should not be directly attributed to mind-wandering, just as the lower hit rates observed in the triple-task condition should not be directly attributed to resource depletion. Although both mechanisms remain theoretically plausible, the present data do not allow us to distinguish them directly. In particular, the absence of a clear task-load effect on sensitivity suggests that the task-load manipulation primarily influenced the criterion used to respond to the EV signal, rather than producing a strong change in the ability to discriminate the signal itself.

Subjective workload showed a partial dissociation from performance. In Study 1, perceptual salience did not modulate NASA-TLX scores, despite clear improvements in hits and false alarms, suggesting that perceptual facilitation can enhance vigilance performance without reducing perceived task demand. In contrast in Study 2, both perceptual salience and task load affected NASA-TLX scores: low salience and higher task load were experienced as more demanding, indicating that subjective workload is sensitive to global task demands.

However, and as mentioned, despite these performance effects, none of the manipulations influenced the vigilance decrement over time. Performance declined steadily across all conditions, suggesting the involvement of additional factors beyond perceptual salience or task load. Thus, the solid pattern of data observed across the three data sets is difficult to reconcile with current resource depletion hypothesis or mind-wandering hypothesis of vigilance.

A central finding across the present work is that neither perceptual salience nor task load robustly modulated the vigilance decrement over time-on-task. In Study 1, the high-

salience condition reduced the decrement in hits, but this pattern was not mirrored by a comparable modulation of sensitivity. In Study 2 and Study 3, the analyses of A' and B'' showed significant changes across blocks, but no significant interactions between block and perceptual salience, block and task load, or block, perceptual salience, and task load. Thus, the manipulations affected the overall level of discrimination and response criterion, but they did not substantially alter the temporal trajectory of the EV decrement.

This pattern has important implications for theories of vigilance decrement. First, it shows that a decrease in hit rates should not automatically be interpreted as a loss of sensitivity. Changes in response criterion can produce similar patterns in hit and FA rates without necessarily implying that participants became less able to discriminate signals from non-signals. Second, the findings suggest that models based solely on overload or underload may be insufficient to explain the full pattern of EV performance. Task demands may influence both signal discriminability and response strategy, and these processes do not necessarily change in the same way over time-on-task (Thomson et al., 2015, 2016).

An alternative interpretation of this pattern is offered by expectancy-based accounts of vigilance, which emphasize adjustments in response criterion rather than changes in perceptual sensitivity over time. In modern vigilance tasks characterized by low signal probability, such as the executive vigilance trials of the ANTI-Vea, performance decrements may be strongly reflected in shifts toward a more conservative response bias, sometimes without a corresponding decrease in sensitivity (Gyles et al., 2023; Roman-Caballero et al., 2023; Thomson et al., 2016). From this perspective, participants may progressively adapt their response strategy to the rarity of critical signals, resulting in fewer responses to both signal and non-signal trials. This account provides a parsimonious explanation for why the

vigilance decrement persisted across conditions even when perceptual salience and task load changed overall performance.

At the same time, the persistence of the vigilance decrement across conditions leaves open the question of which mechanisms drive performance decline over time. One relevant candidate is mind-wandering. Numerous studies have shown that mind-wandering increases as time-on-task progresses, while task performance tends to decrease (Smallwood, 2010; Smallwood & Schooler, 2006, Thomson, et al., 2015). According to resource-control theory, attentional resources do not necessarily disappear over time; rather, the allocation of resources between task-related and task-unrelated activity is regulated by cognitive control (Thomson et al., 2015). Over time, the ability to maintain task focus and suppress internal distractions may weaken, allowing attention to drift away from task goals. However, the present studies did not directly measure mind-wandering during task performance. Therefore, mind-wandering should be considered a plausible mechanism for future research rather than a demonstrated explanation of the present task-load effects. The same caution applies to resource depletion: although the overload framework provides a useful account of how high task demands may impair vigilance, the current results do not show that the triple-task condition reduced sensitivity through a depletion of attentional resources. Thus, the present findings are best interpreted as showing that perceptual salience and task load affect different components of EV performance, while the mechanisms underlying the time-on-task decrement remain to be directly tested.

Implications, limitations and Future Directions

The present findings have practical implications for the design of monitoring systems and tasks requiring sustained attention. Vigilance decrement can lead to a wide range of

consequences, from relatively mild effects such as missing a critical signal to severe outcomes including fatal car crashes (Baysari et al., 2008; Edkins & Pollock, 1997; Wundersitz, 2019). Increasing the perceptual salience of critical signals reliably improved detection performance and sensitivity without necessarily increasing overall task demands. In addition, introducing moderate levels of task load may help sustain attentional engagement in monotonous or low-stimulation environments. However, the present findings also indicate that improving raw detection performance does not necessarily mean that the vigilance decrement has been eliminated. Even when perceptual salience improved sensitivity and task load modified response bias, neither manipulation robustly prevented the time-on-task decrement. Therefore, applied interventions should consider not only whether signals are detected more often, but also whether users are changing their response criterion over time.

The findings regarding task load also have applied relevance. The dual-task condition was associated with the least conservative response criterion, whereas the triple-task condition promoted the most conservative criterion. Thus, moderate task demands may increase participants' willingness to respond to vigilance signals, but this should not be interpreted as evidence of improved sensitivity or as direct evidence for an optimal level of attentional resources. In applied settings, reducing misses by encouraging a less conservative criterion may also increase the risk of FA, so the balance between sensitivity and response bias should be carefully considered.

The ANTI-Vea is available as an online, open-access platform (<https://anti-vea.ugr.es/>), which facilitates its implementation across a wide range of experimental and applied contexts. A key strength of the task is its modular structure, which allows different subtasks to be combined or isolated depending on the research question, enabling flexible

1113 manipulation of perceptual, executive, and attentional demands. This flexibility makes the
1114 ANTI-Vea particularly suitable for studying vigilance in applied domains such as sleep
1115 deprivation, fatigue, circadian rhythms, or workload management, where task demands can
1116 be adapted to reflect real-world operational constraints.

1117 Moreover, its online format allows for large-scale data collection and remote testing,
1118 which is especially valuable for human factors research aiming to assess sustained attention
1119 under ecologically relevant conditions. As such, the ANTI-Vea provides a versatile
1120 methodological framework that can be readily adapted to bridge laboratory-based vigilance
1121 research with applied and real-world settings

1122 An important limitation is that the contradictory findings across studies may partly
1123 stem from differences in experimental design, as Study 1 employed a mixed-participant
1124 design whereas Study 2 used a within-participant design. Mixed or between-participant
1125 designs are less sensitive to subtle effects and more susceptible to inter-individual variability,
1126 which may be particularly relevant when attempting to replicate prior work that
1127 predominantly used within-subjects approaches (Charness et al., 2012).

1128 Another limitation of the present work is that task load was manipulated through
1129 qualitatively different task configurations rather than equidistant levels along a single
1130 quantitative dimension. Accordingly, the single-, dual-, and triple-task conditions differed not
1131 only in the nominal “level” of load, but also in the specific cognitive demands they imposed,
1132 such as response inhibition or increased working memory load. Within this framework, the
1133 inverted-U-shape performance function is offered as an interpretation of the patterns observed
1134 under these particular task-load conditions. Although this pattern is consistent with models
1135 positing optimal levels of attentional engagement, its generalization should be understood as

referring to this specific class of task configurations. Consequently, future studies should examine whether similar performance patterns and vigilance dynamics emerge under alternative single-, dual-, and triple-task conditions that involve different combinations of cognitive processes.

An additional limitation, the perceptual-salience manipulation was based on three discrete levels of vertical displacement. This design was useful for comparing conditions of different perceptual difficulty and was consistent with the preregistered analyses of Studies 1 and 2, which focused on hits, FA, sensitivity, and response bias. However, the present studies were not designed to estimate individual psychometric functions. Although such analyses would provide a more fine-grained characterization of perceptual sensitivity and response bias, they were beyond the scope of the present design and analytic plan. Future research should use psychometric-curve analyses to distinguish more precisely changes in perceptual sensitivity, internal processing noise, lapse rate, guess rate, and response criterion (Gyles et al., 2023; Roman-Caballero et al., 2023).

A further limitation of this work is the absence of direct measures of mind-wandering and cognitive effort, which constrains a full understanding of the mechanisms underlying vigilance decrement. Future research should incorporate self-report, behavioral, or physiological indicators to better characterize these processes and refine explanatory models. Finally, although the final reanalysis enhanced generalizability by including a larger and more diverse sample, data were collected under controlled laboratory settings or instructed home-based protocols, predominantly involving young adult participants. Replicating these findings in applied contexts and across broader populations would be essential to strengthen ecological validity.

1159 **Conclusions**

1160 Overall, results obtained using the ANTI-Vea task offer a sensitive assessment of EV
1161 and demonstrate that perceptual salience and task load influence vigilance performance in
1162 complex, interacting ways. Integrating measures of mind-wandering and cognitive control in
1163 future studies could further advance understanding of how concurrent demands and temporal
1164 dynamics interact to shape attentional control during prolonged tasks.

1165 Key Points

- 1166 • Performance (hits, FA, sensitivity and response bias) in vigilance tasks
1167 typically declines over time, posing risks in contexts that require monitoring
1168 the environment to detect critical stimuli.
- 1169 • This study examined the roles of perceptual salience and task load in the
1170 vigilance decrement using the ANTI-Vea task.
- 1171 • Perceptual salience robustly improved sensitivity, indicating better
1172 discrimination of signal from non-signal trials.
- 1173 • Task load primarily affected response bias: participants adopted the most
1174 conservative criterion in the triple-task condition and the least conservative
1175 criterion in the dual-task condition.
- 1176 • More salient stimuli and a moderate task load (dual-task) were associated with
1177 better performance.
- 1178 • Neither perceptual salience nor task load altered the vigilance decrement over
1179 time.
- 1180 • The findings highlight the importance of balancing perceptual stimulation and
1181 cognitive demands to optimize vigilance.

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